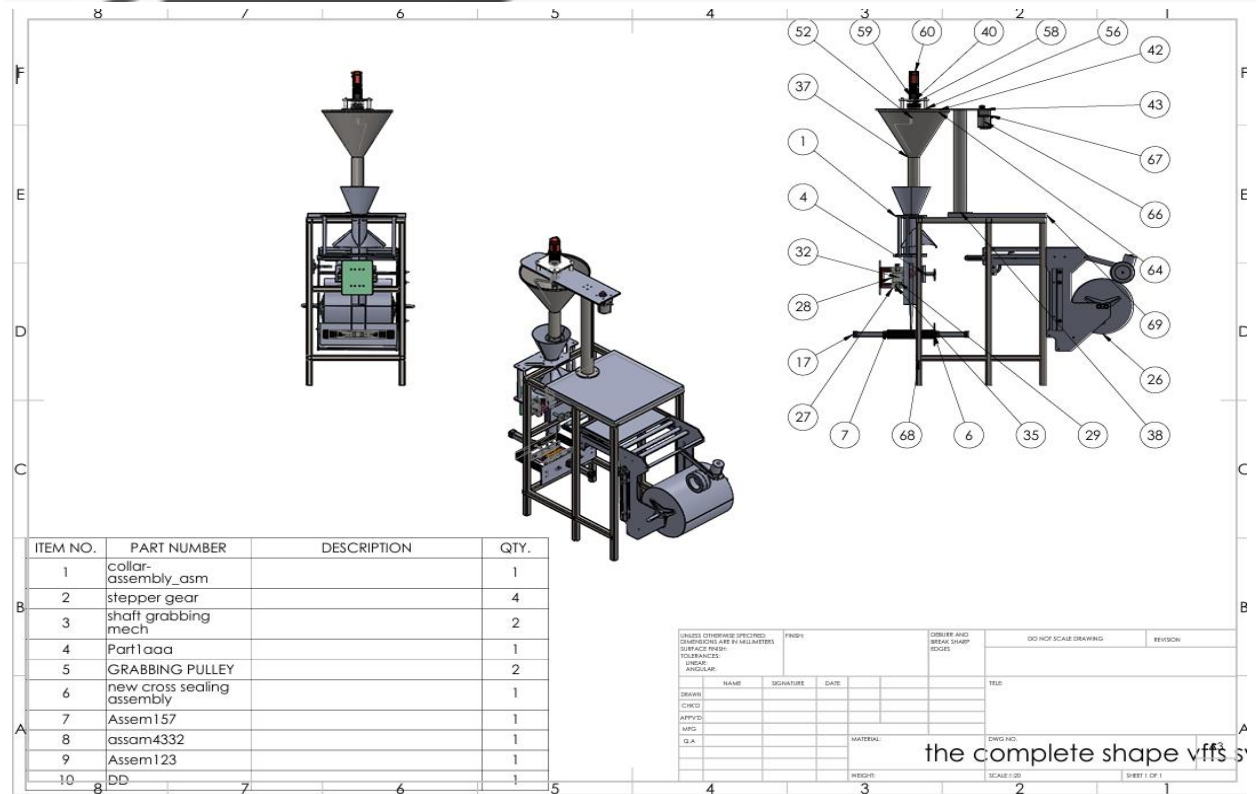
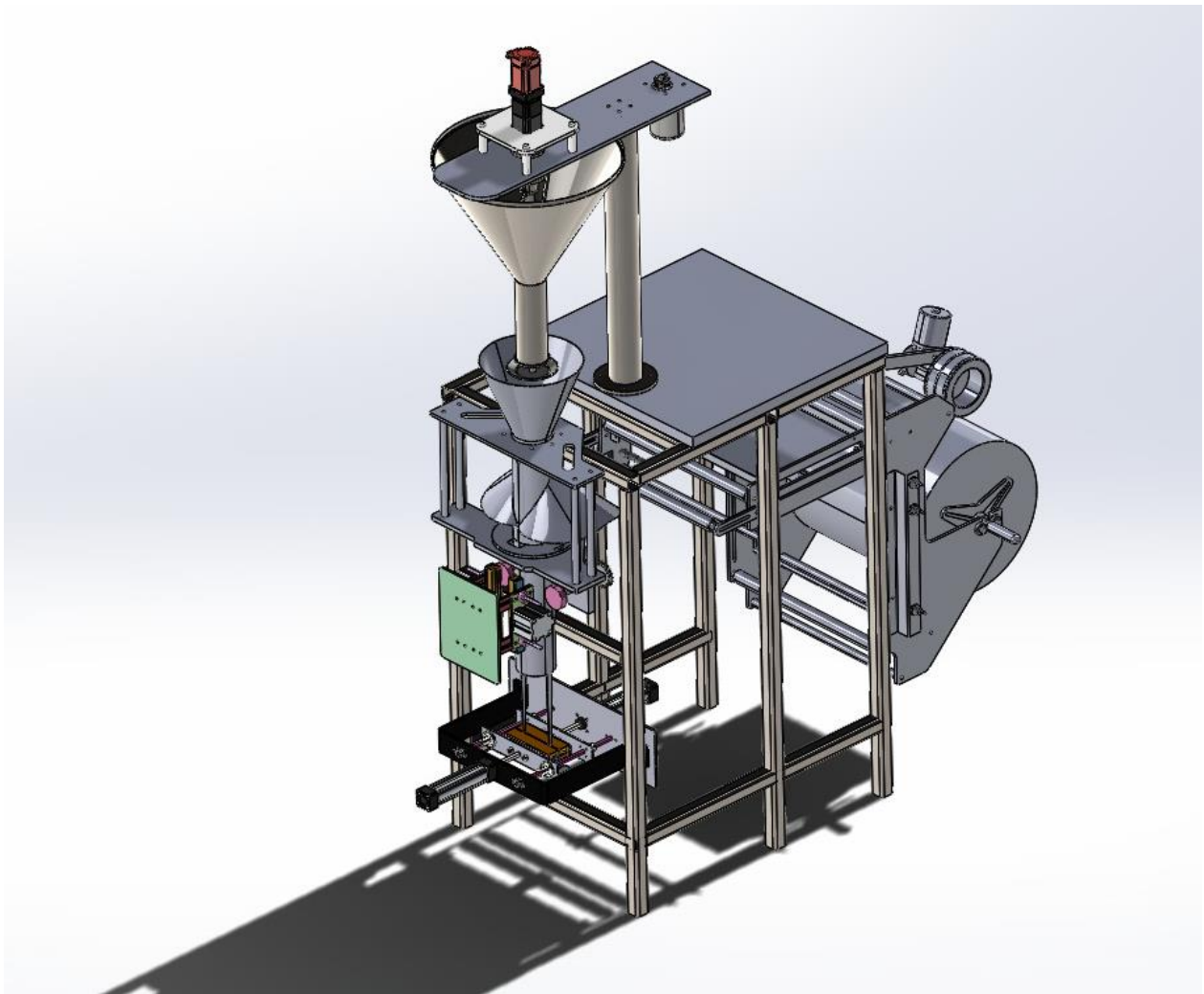


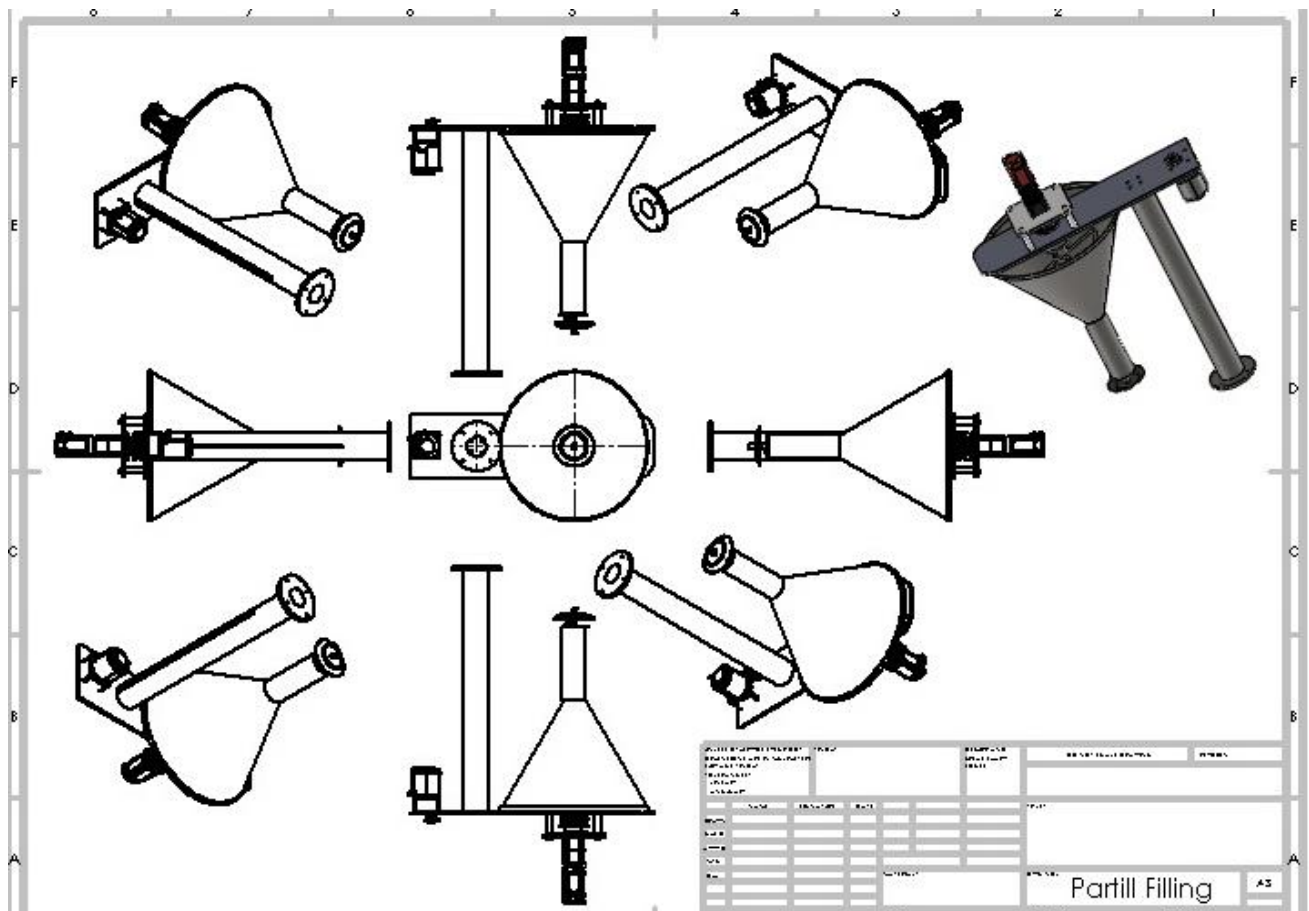
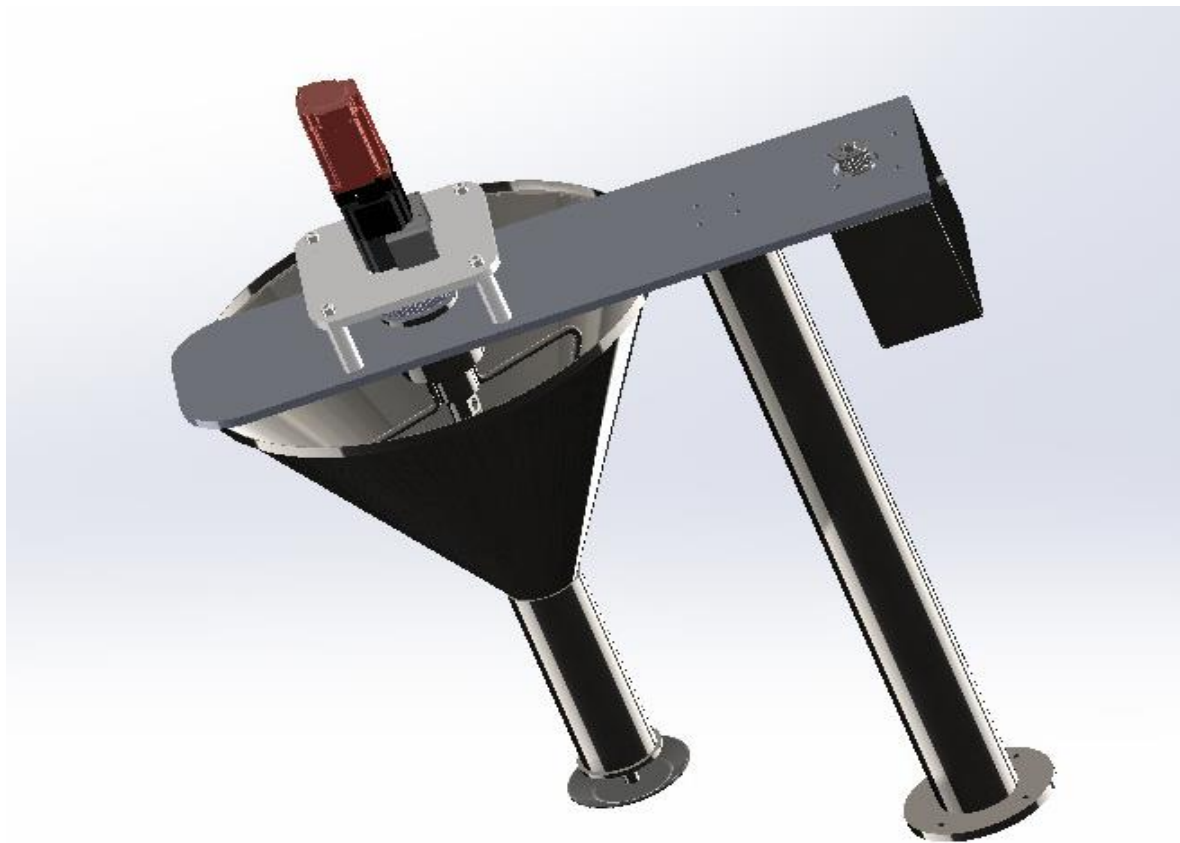
Appendix A: Engineering Drawings (Mechanical)

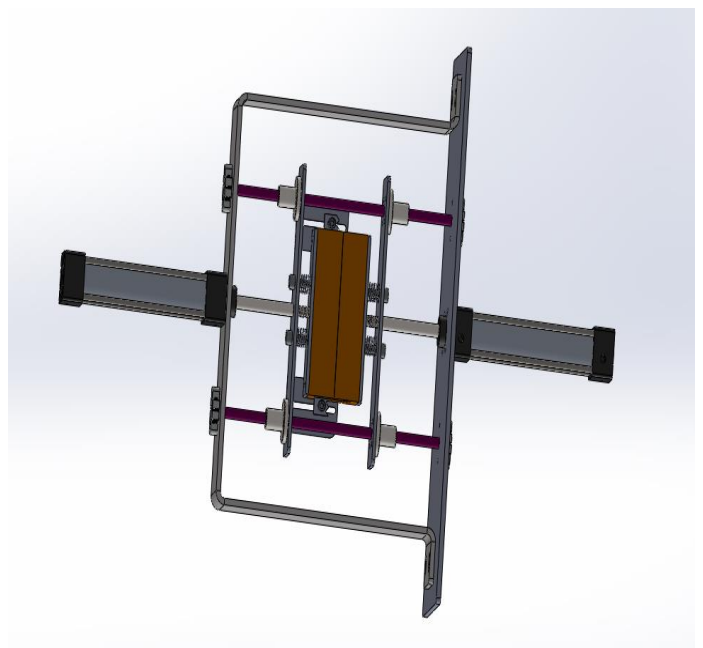
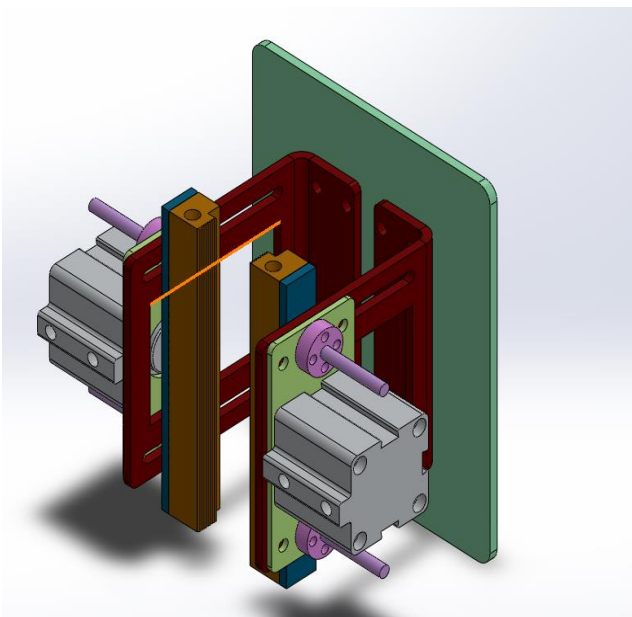
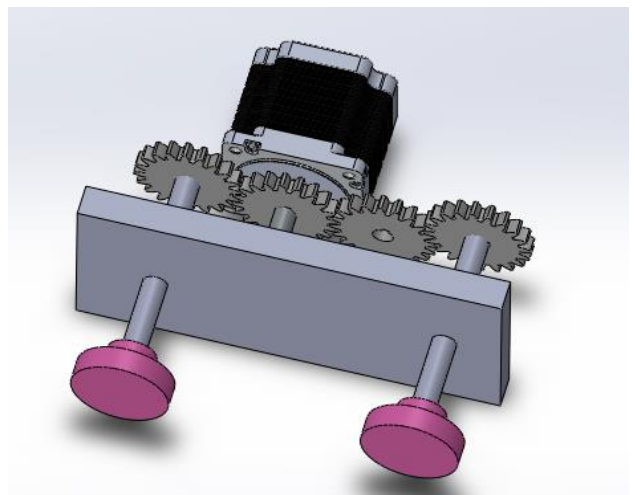
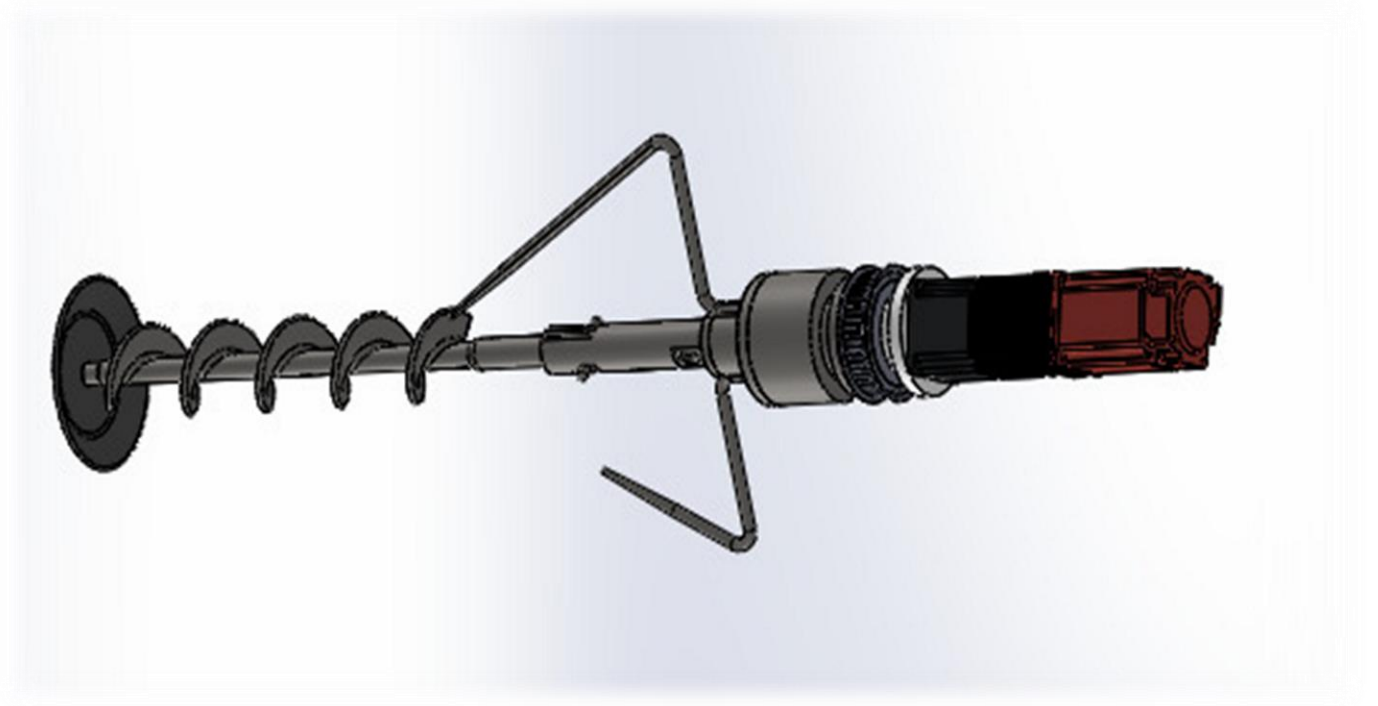
A.1. General Assembly (GA) Drawing.



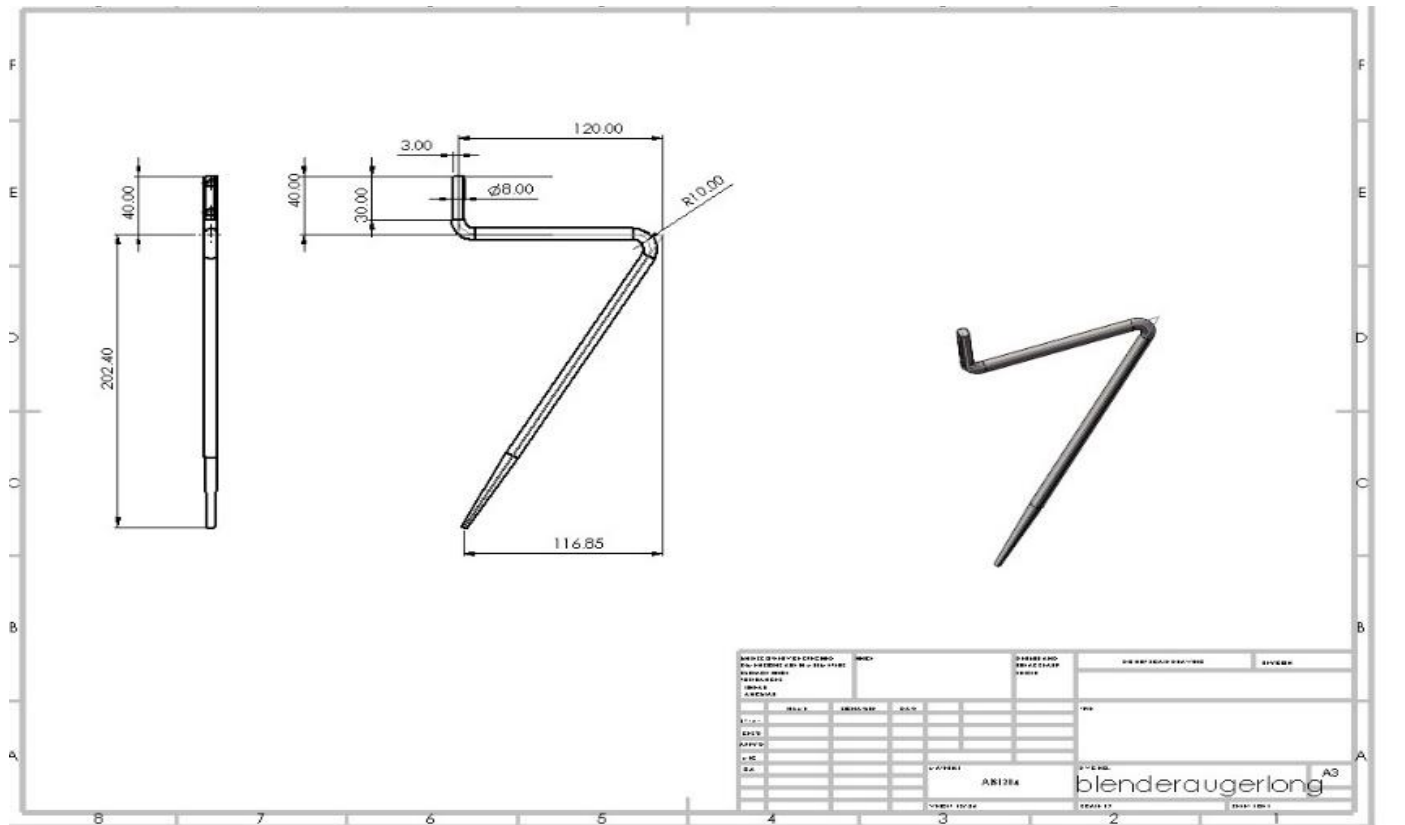
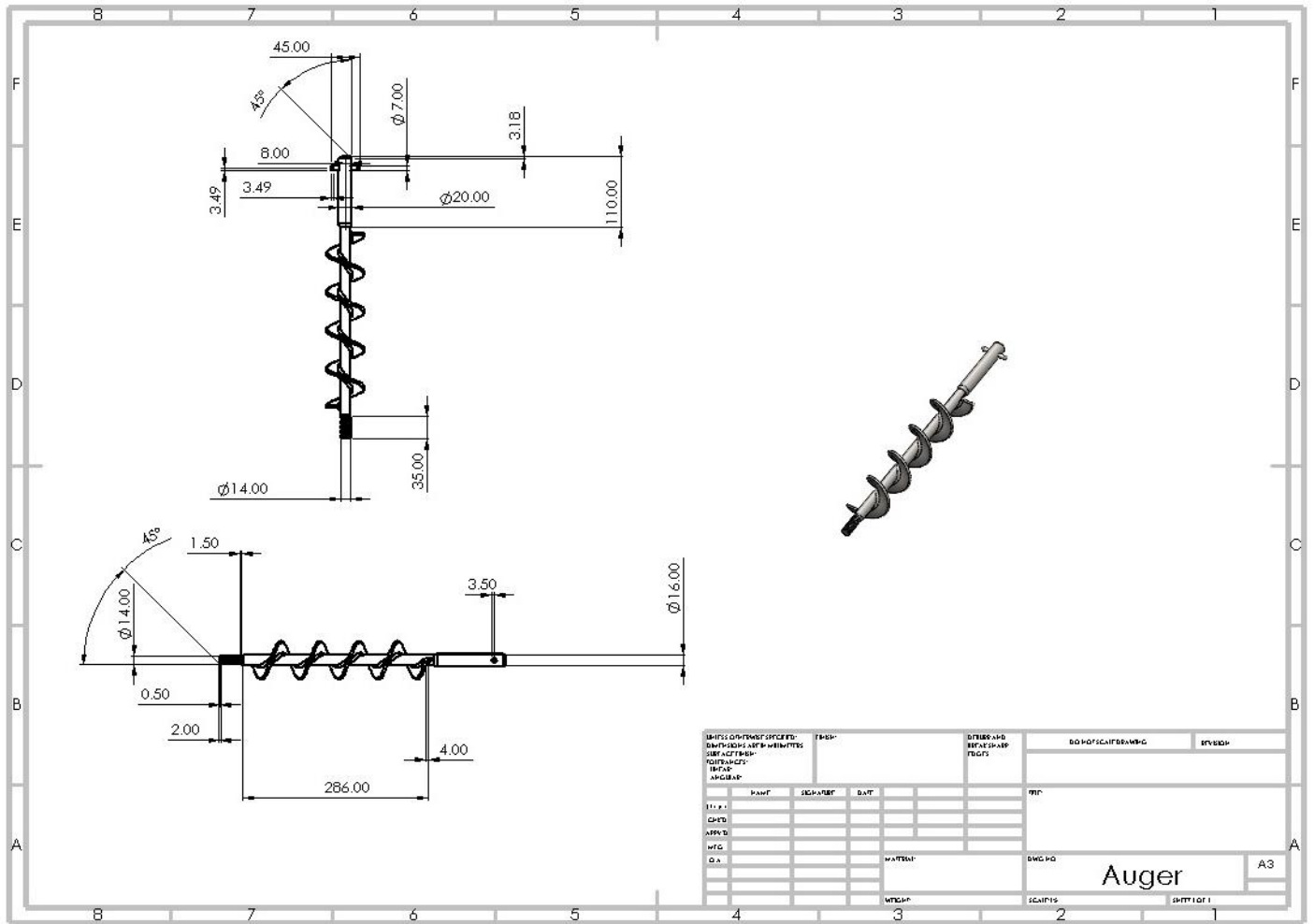


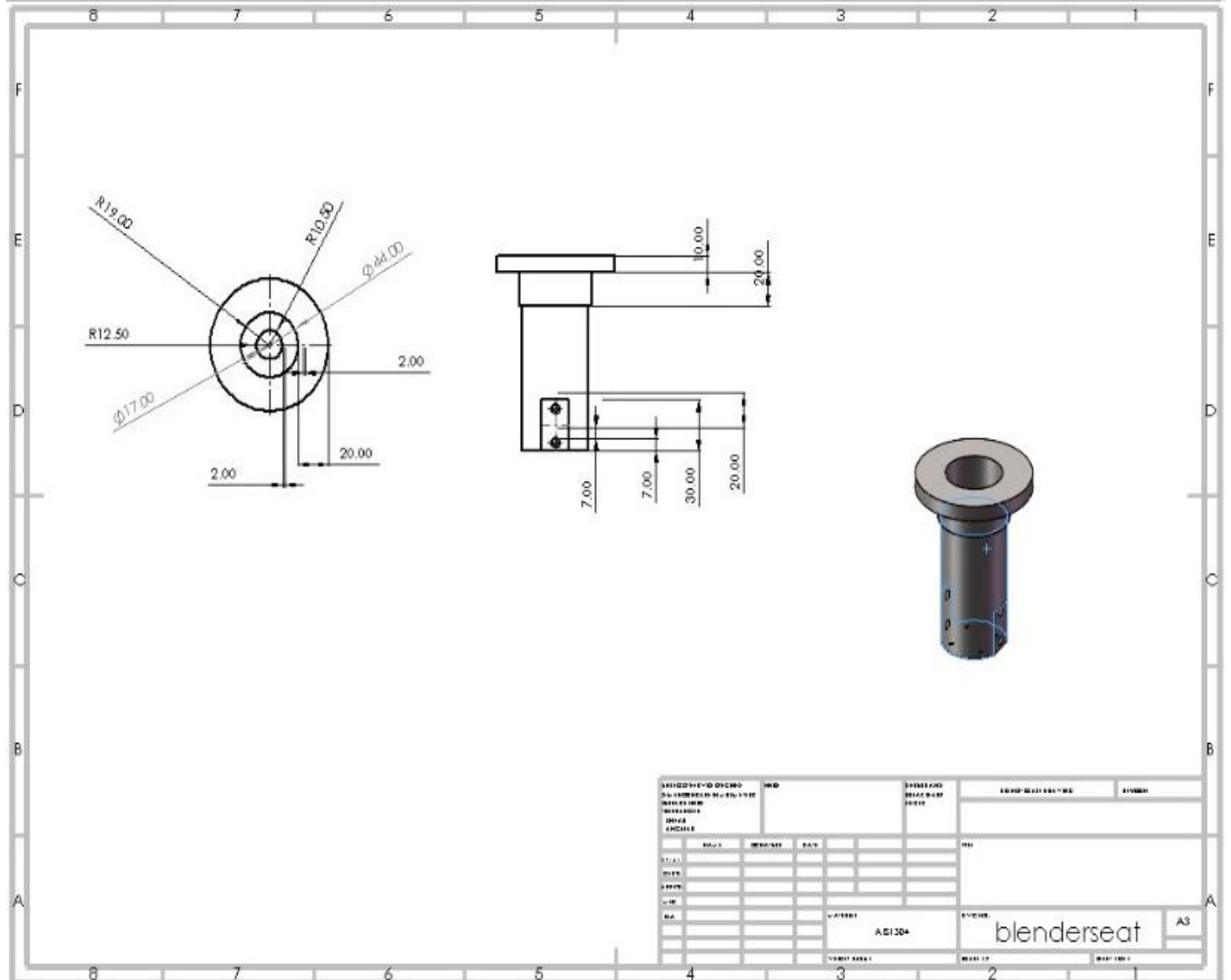
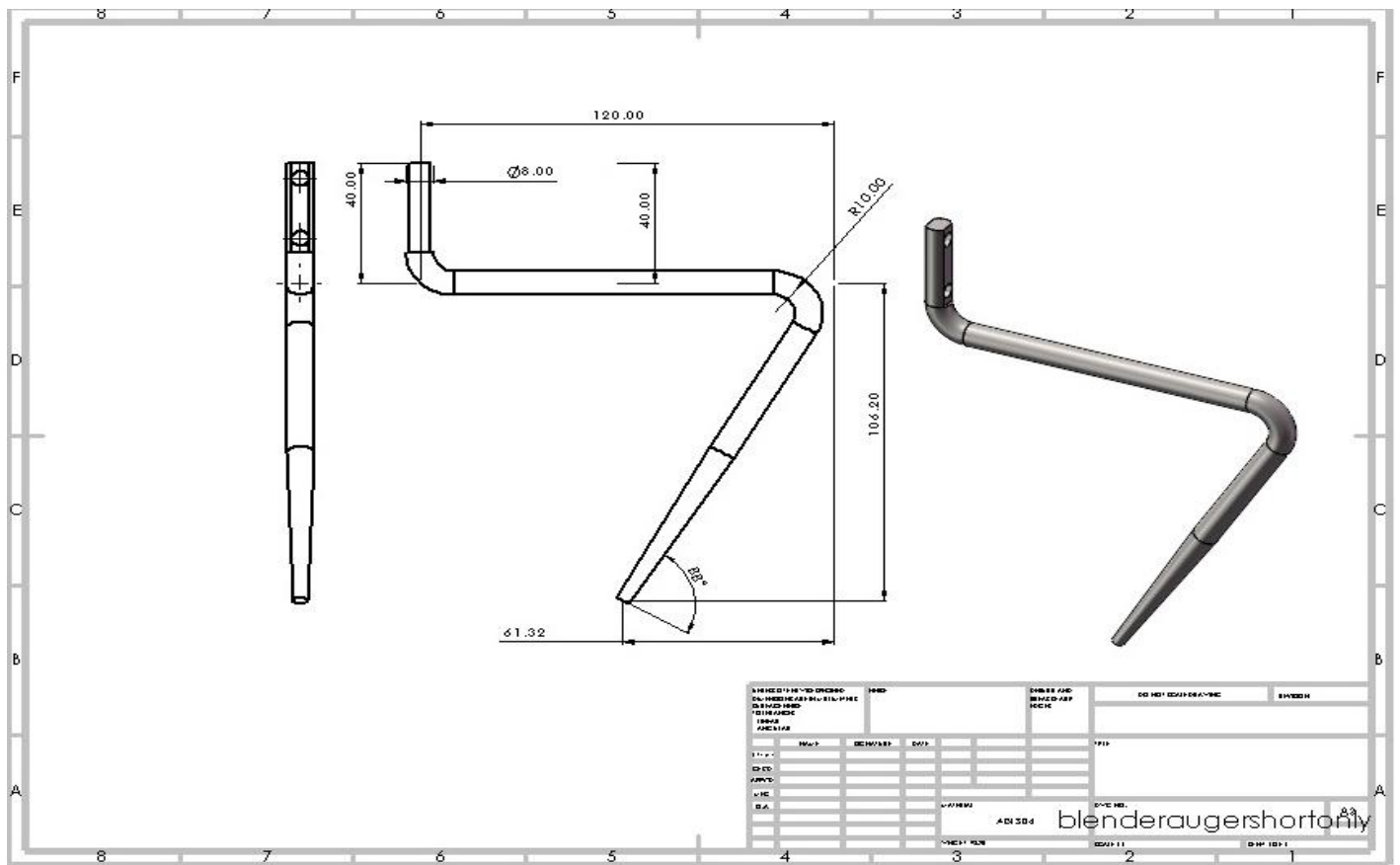
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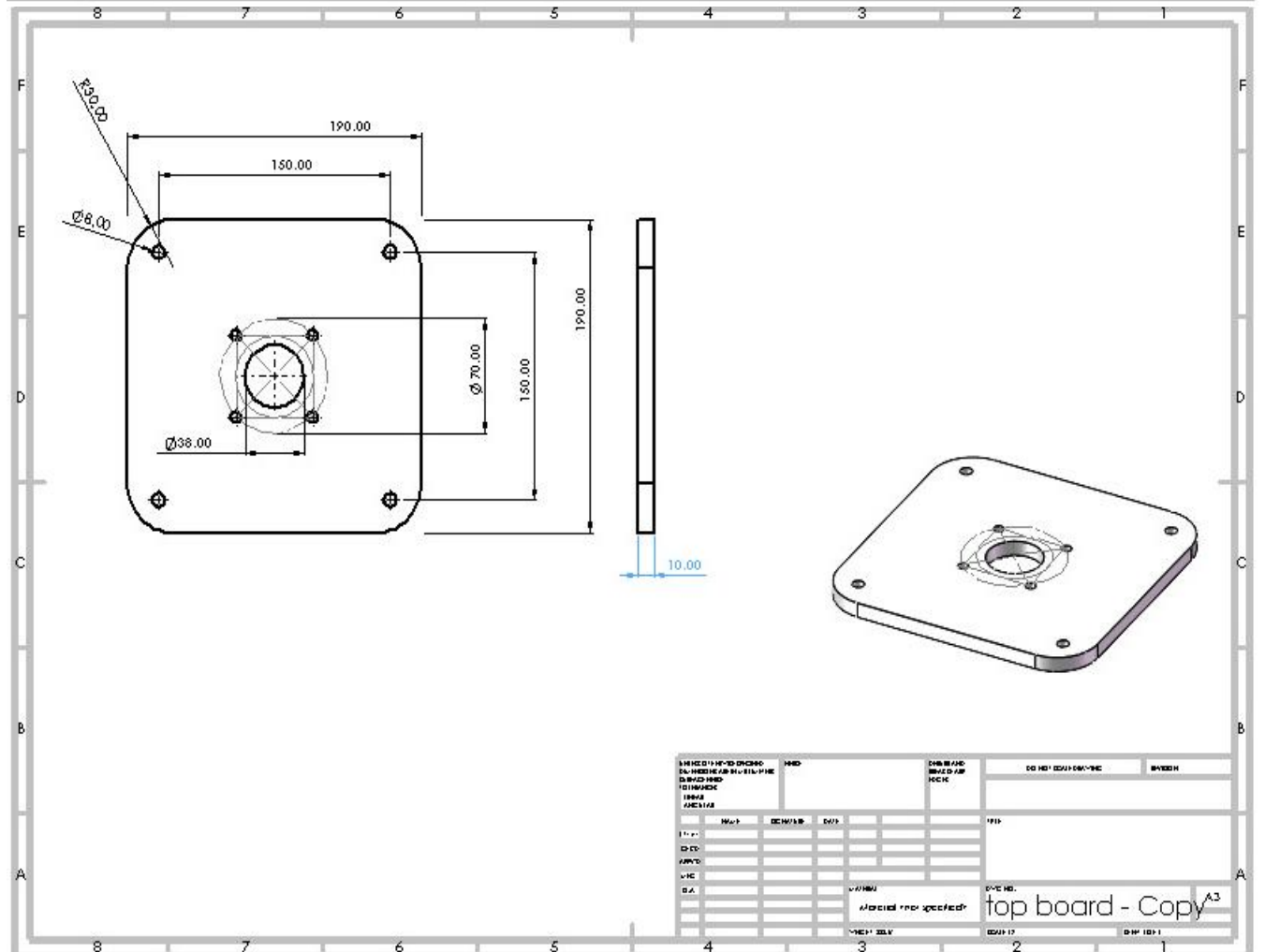
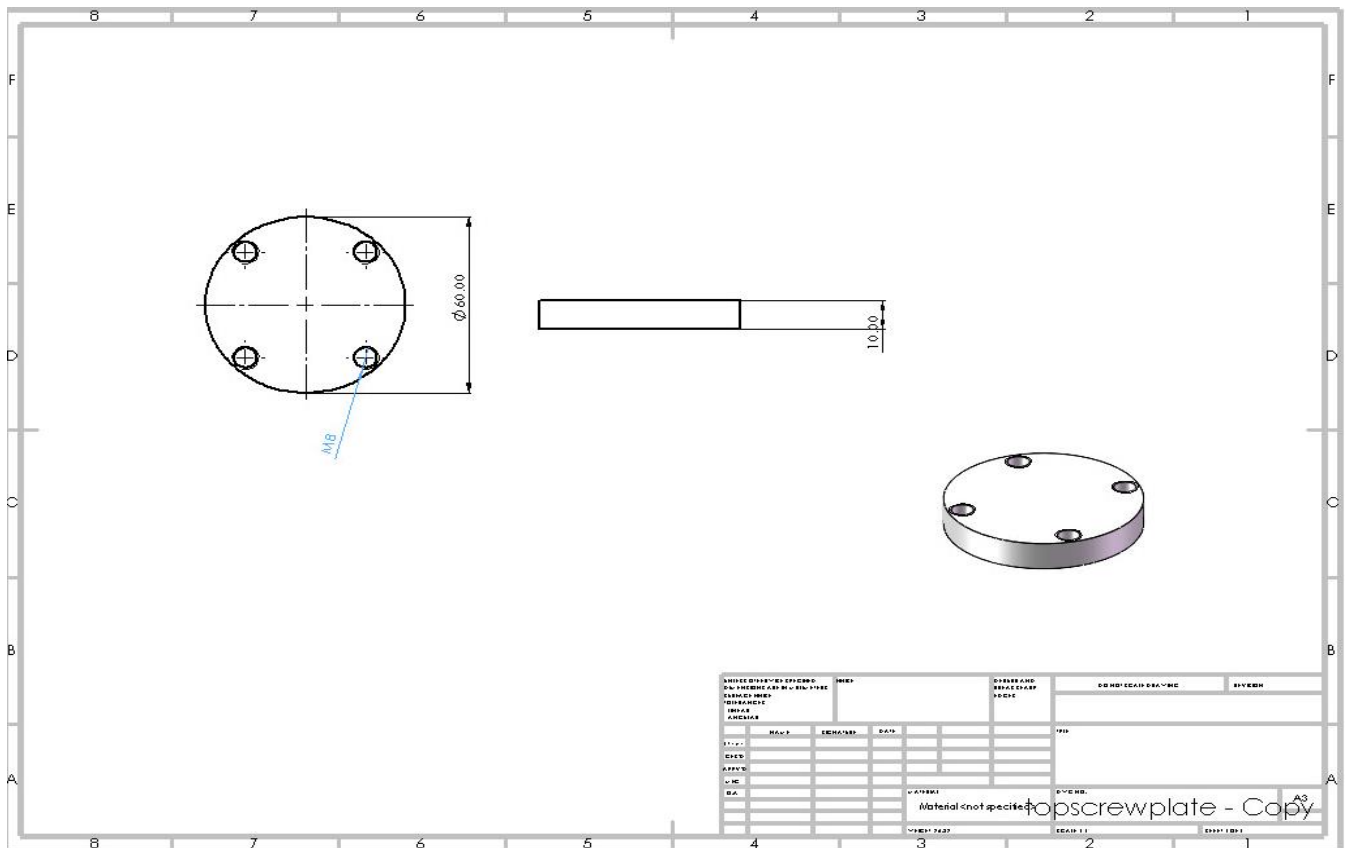


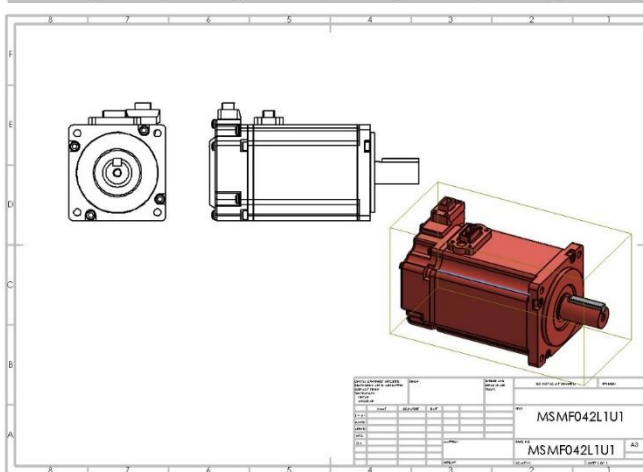
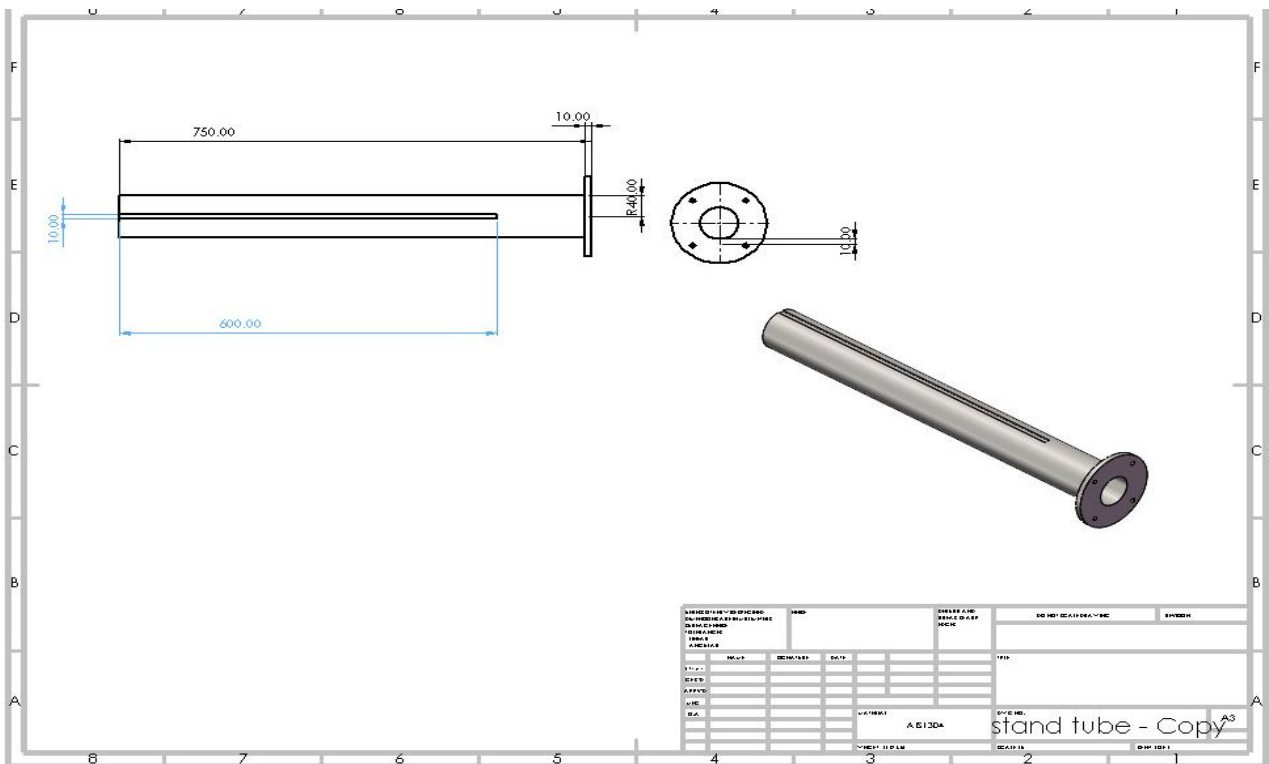


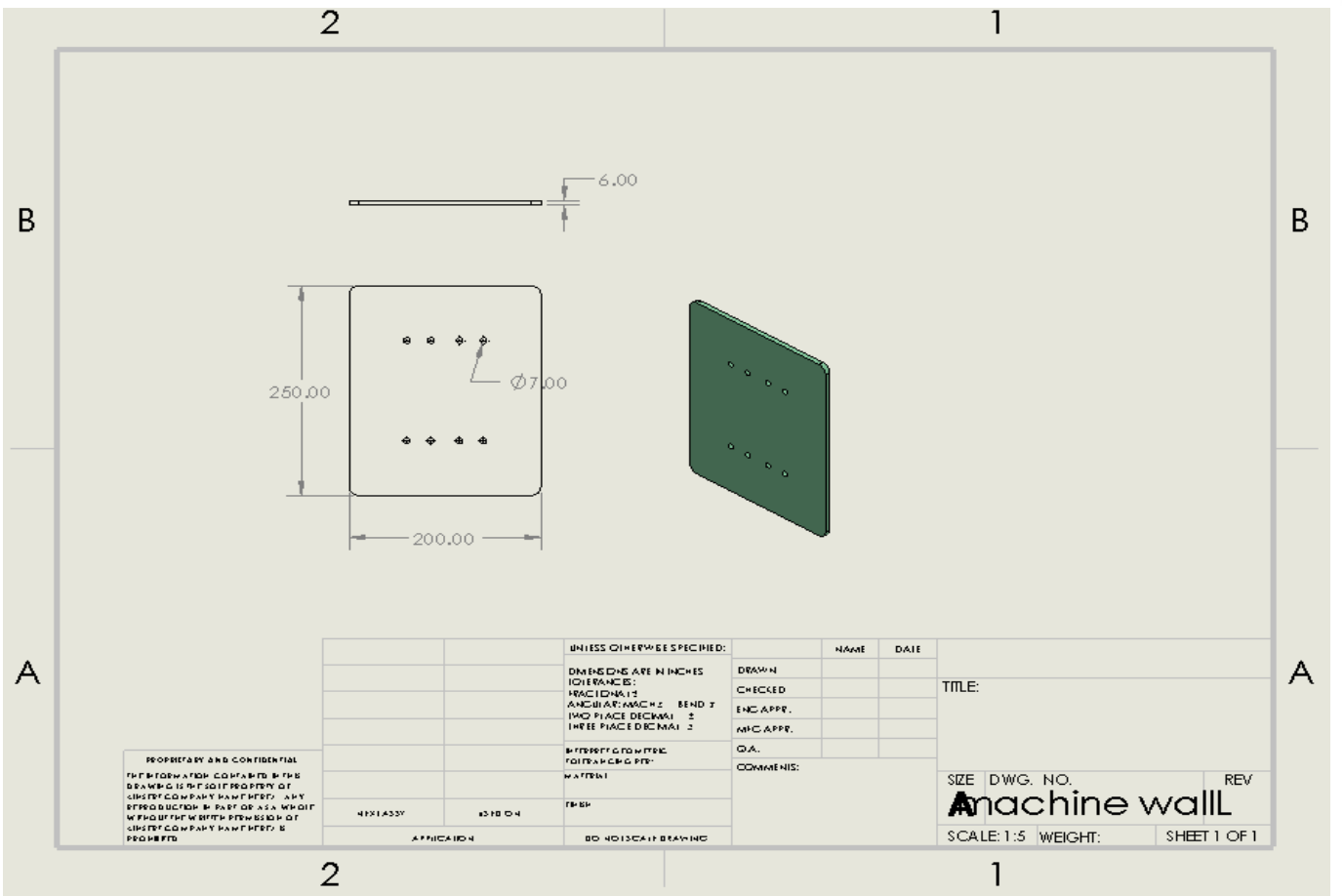
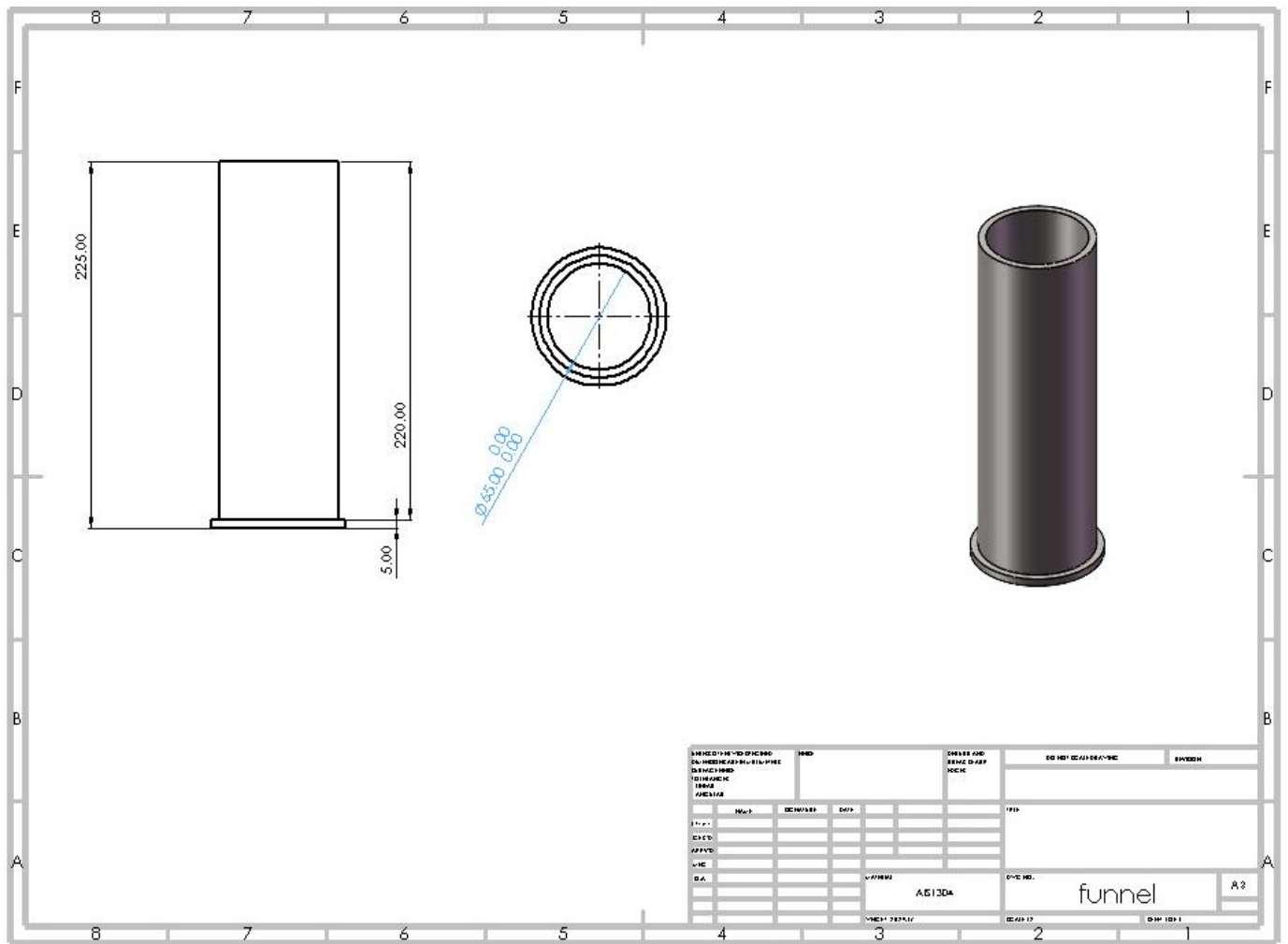
A.3. Detailed Component Drawings.

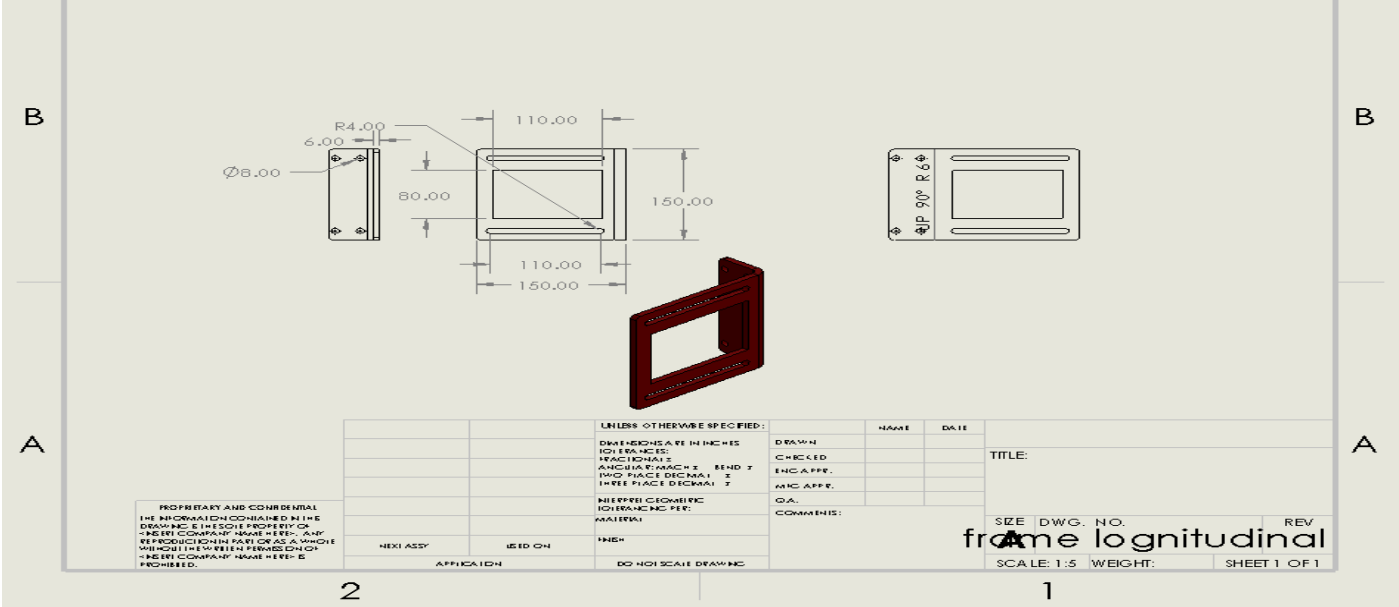
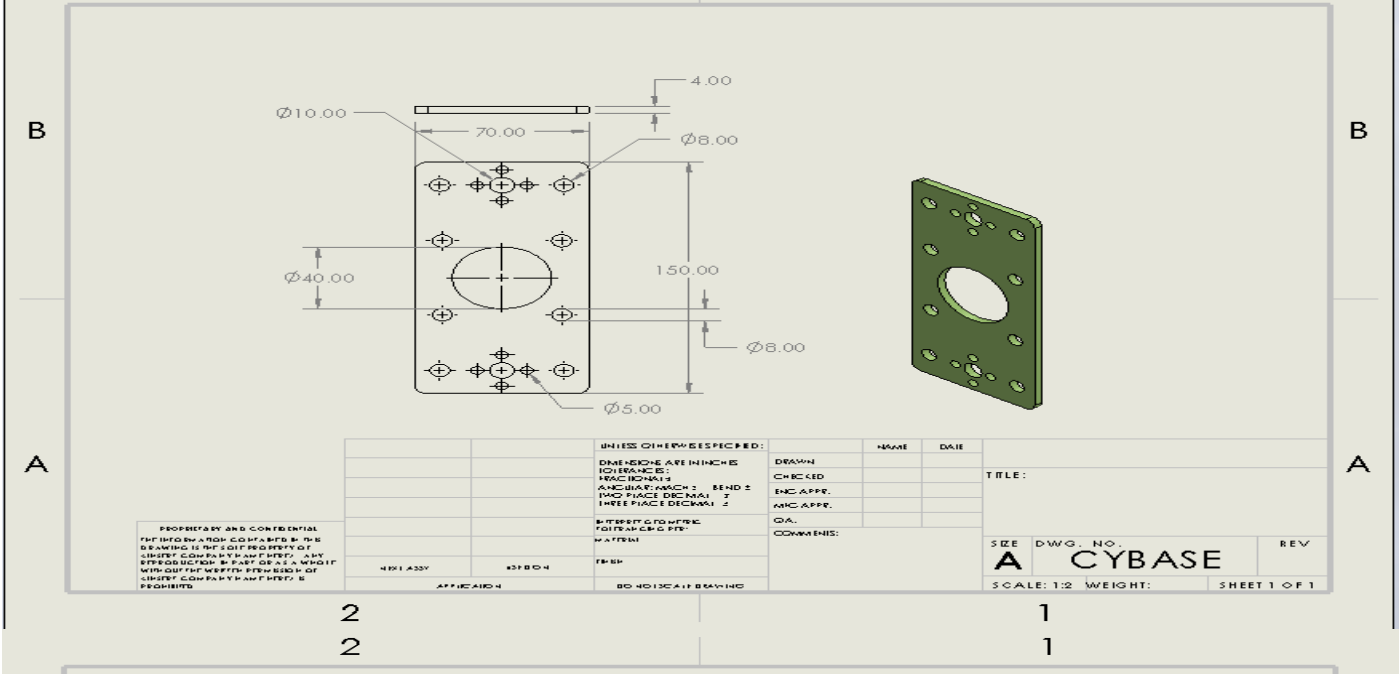
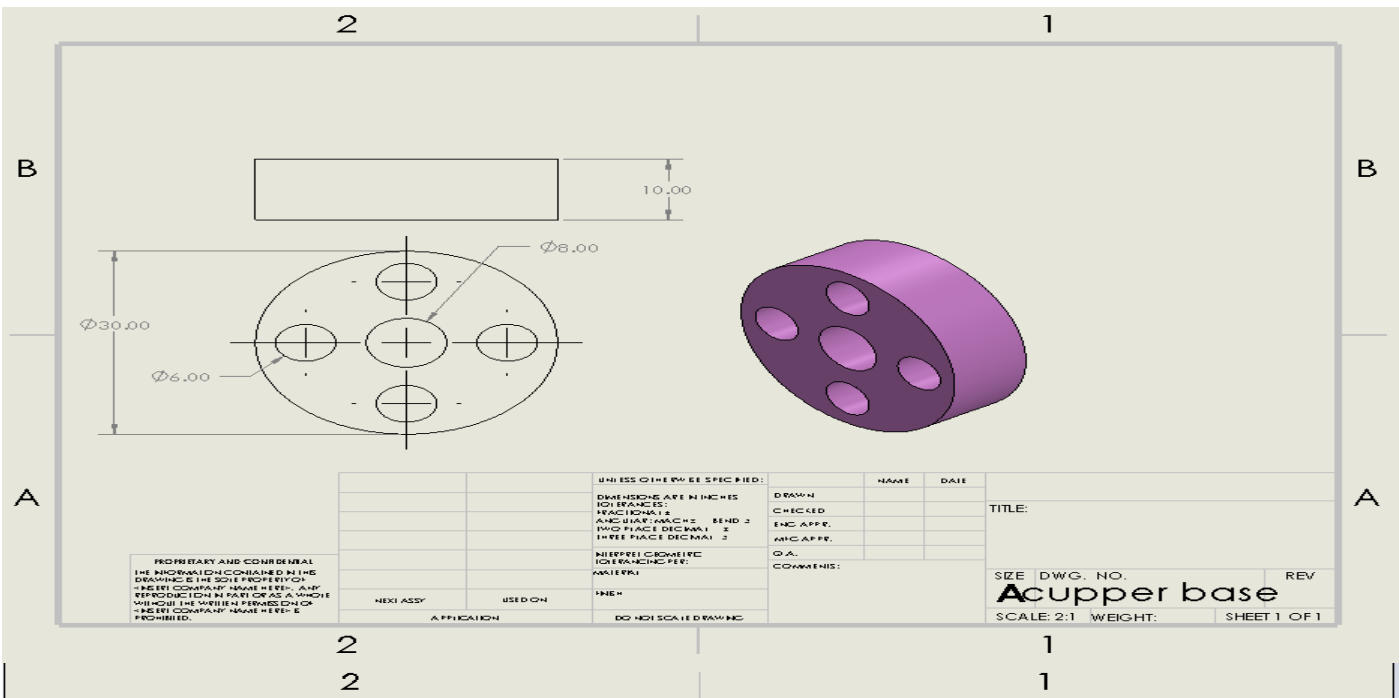


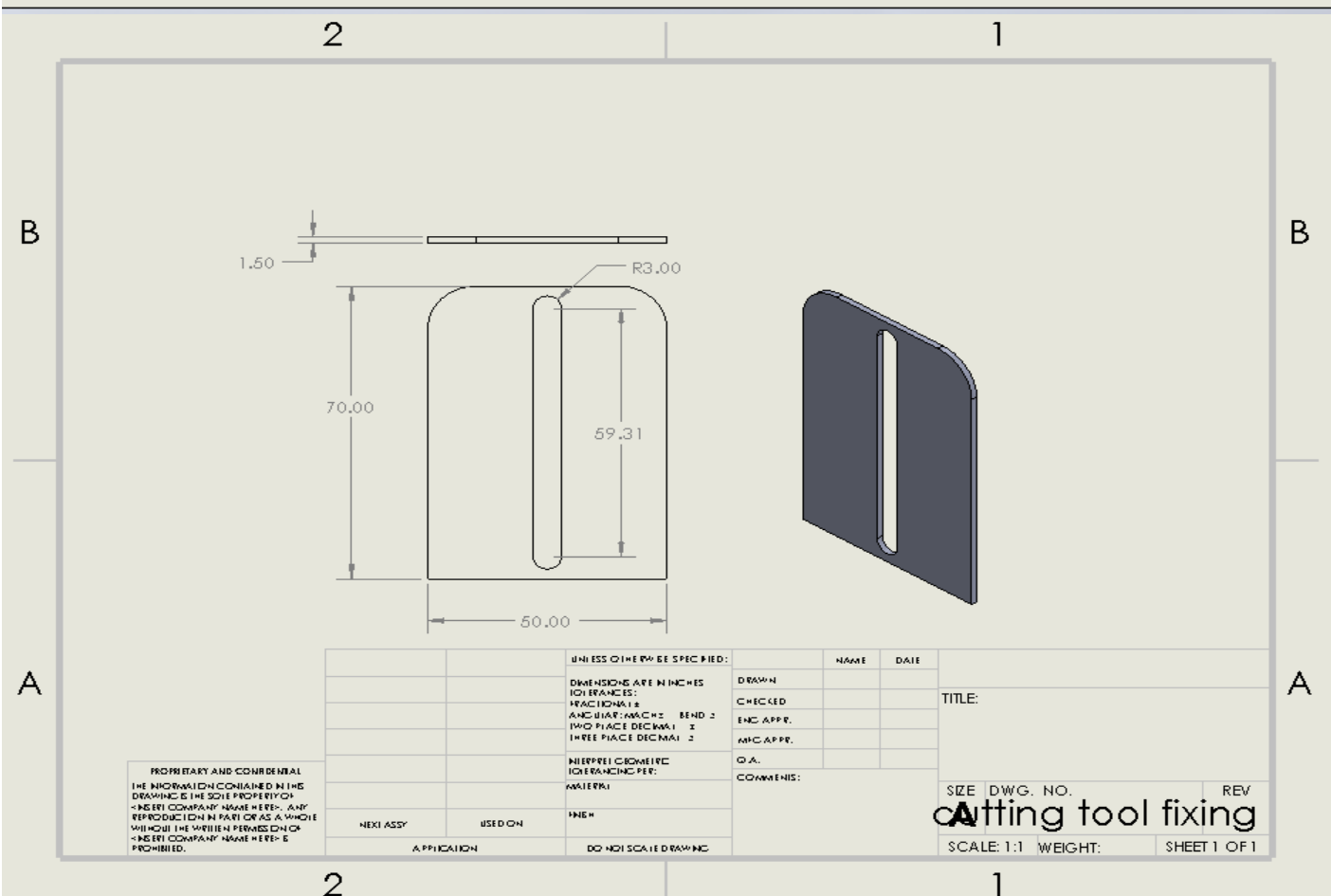
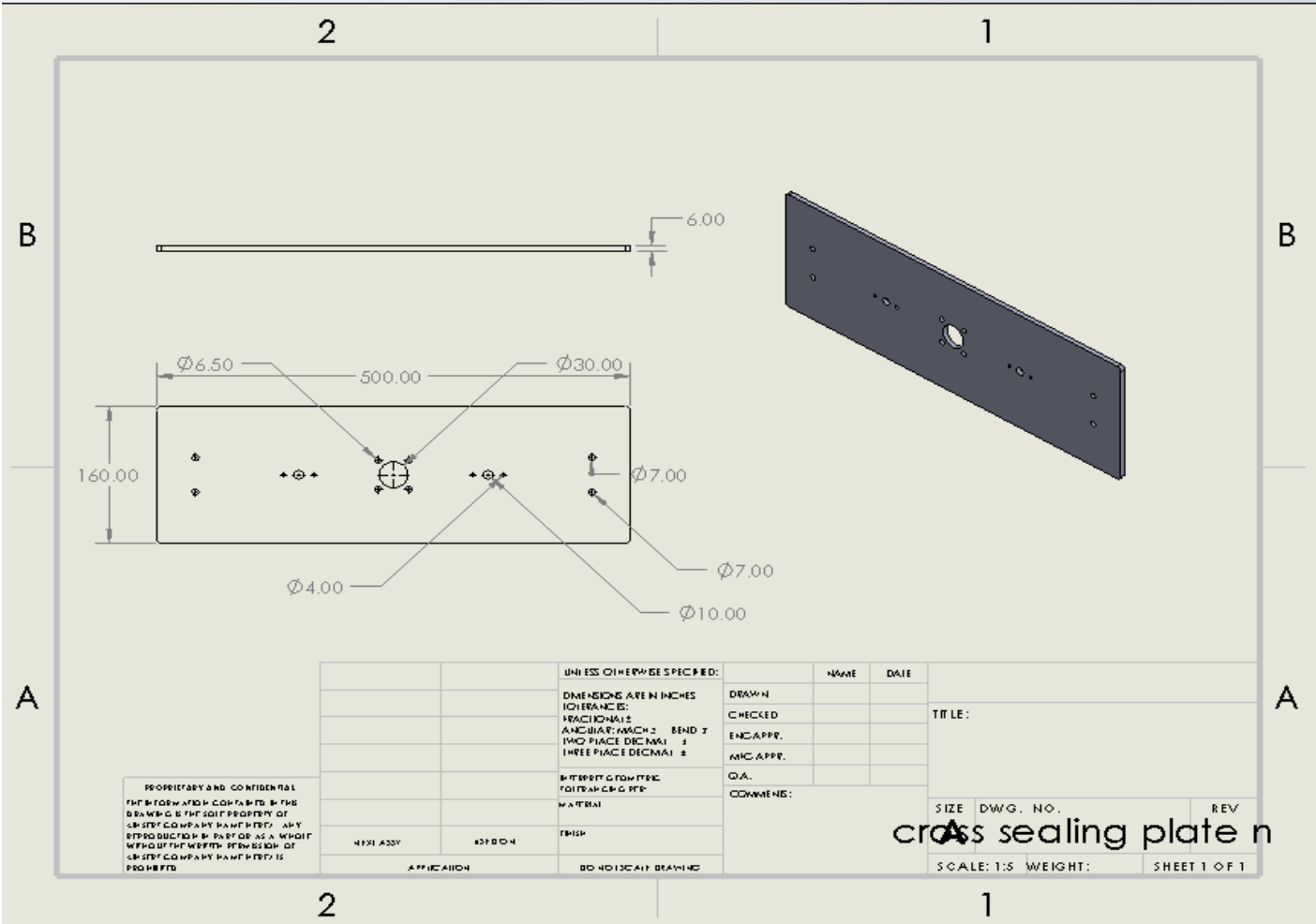












B



- 180.00

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<INSERT COMPANY NAME HERE> IS
PROHIBITED.

SIZE	DWG. NO.	REV.
A	cutting tool	
SCALE: 1:2	WEIGHT:	SHEET 1 OF

1

1

D

DOWN 90° R 6

UP 90° R 6

Ø6.50

- Ø5.0

Ø 10.00

2

DOWN 90° R6

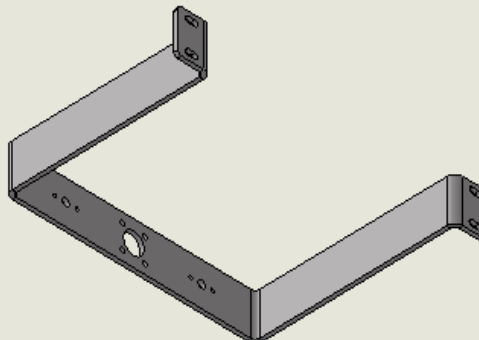
- R4 00

-6.00

C



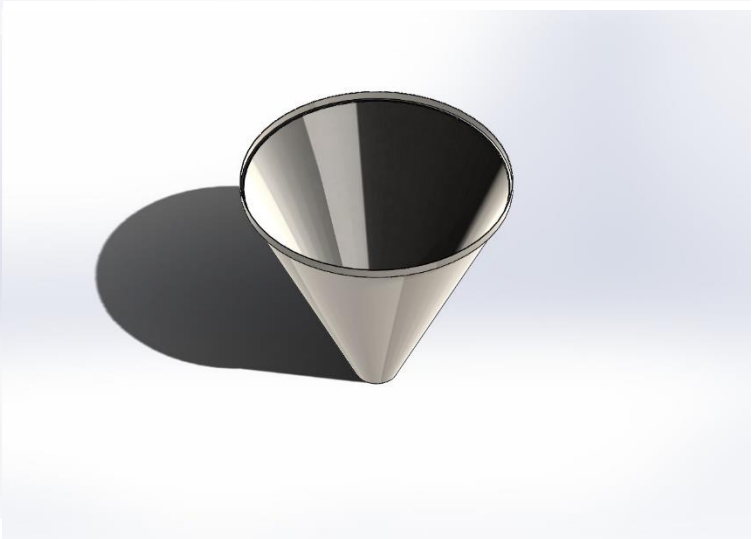
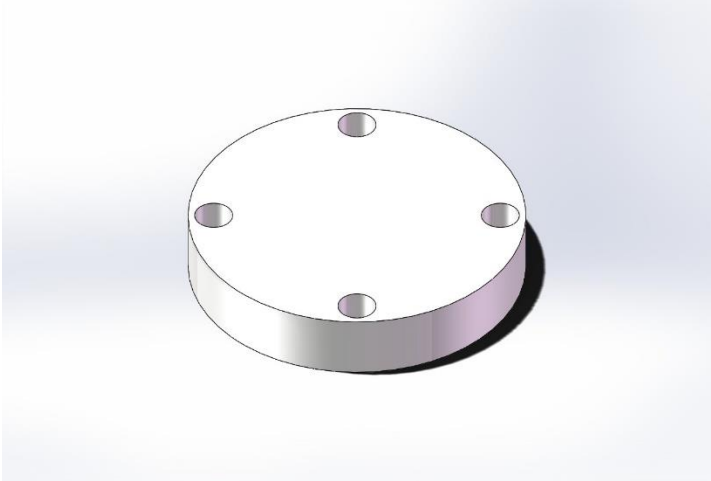
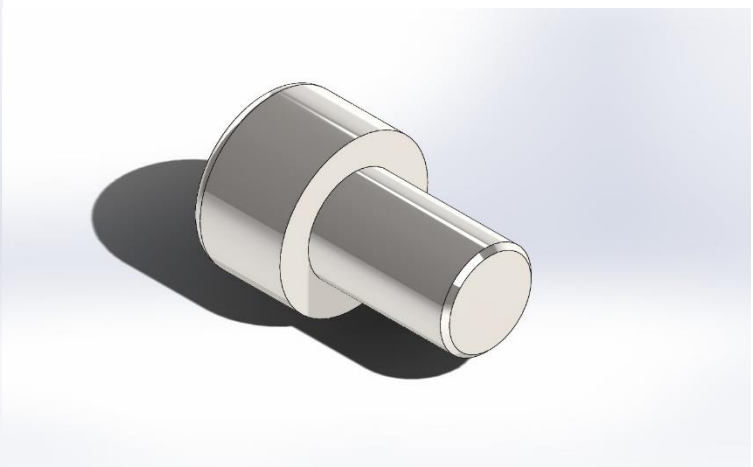
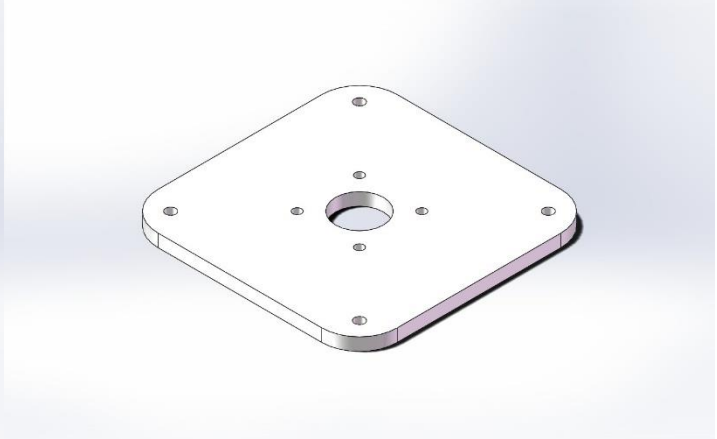
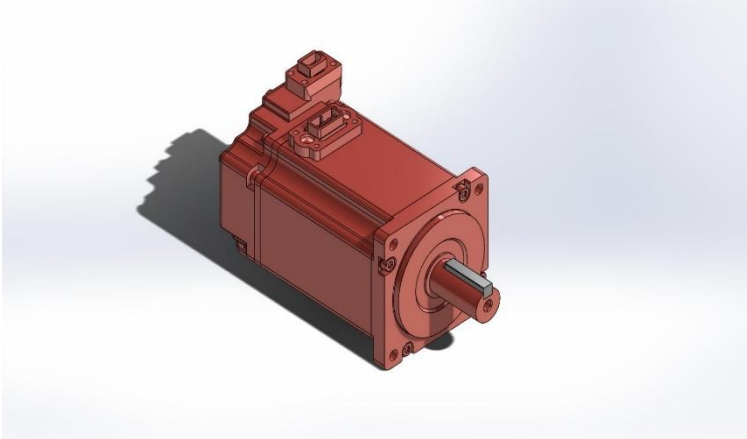
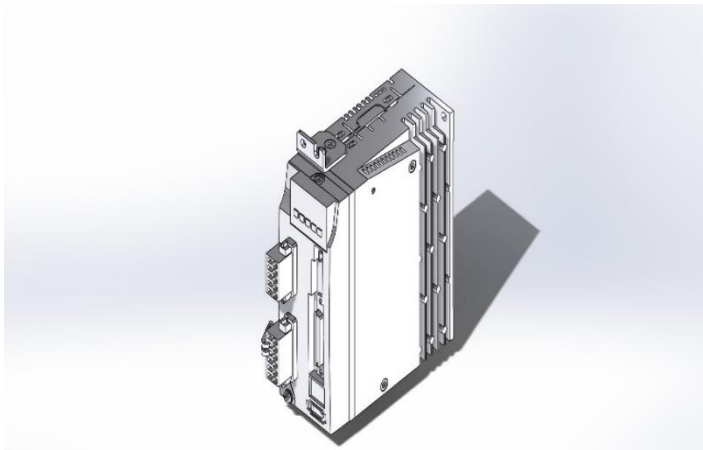
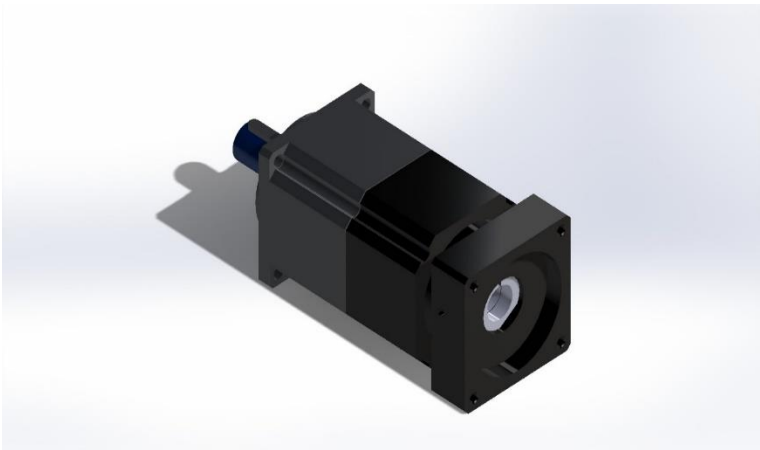
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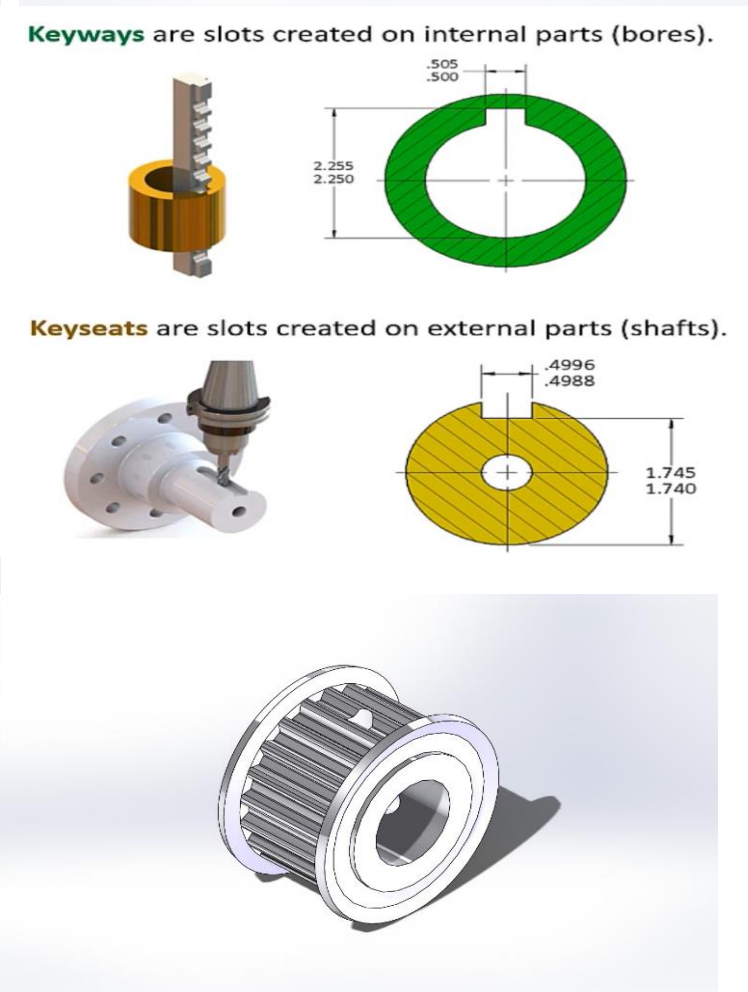
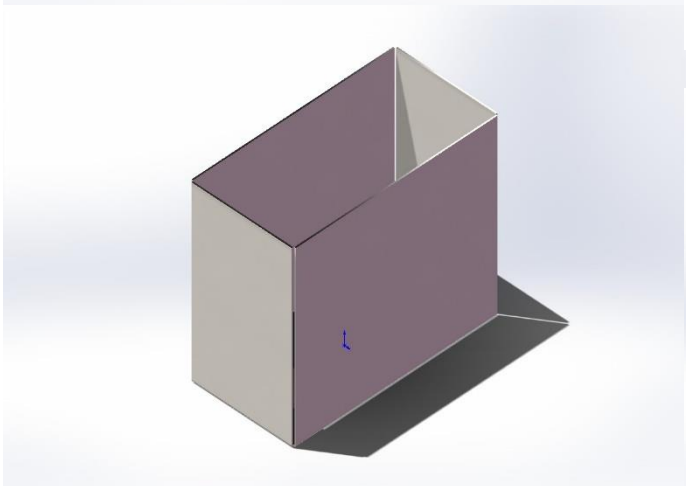
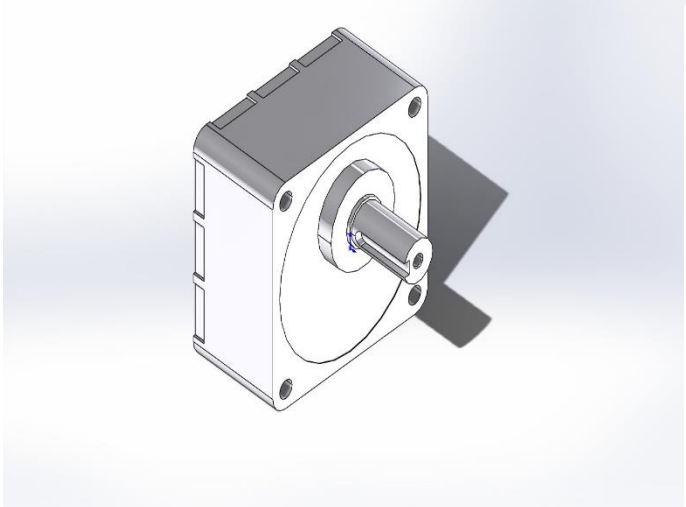
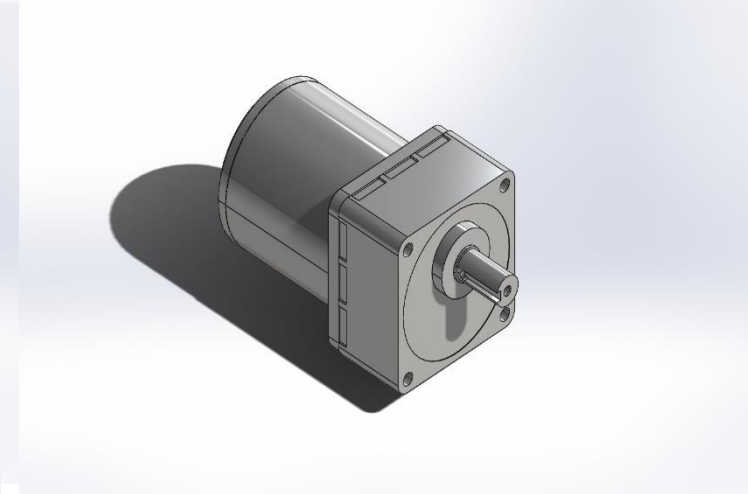
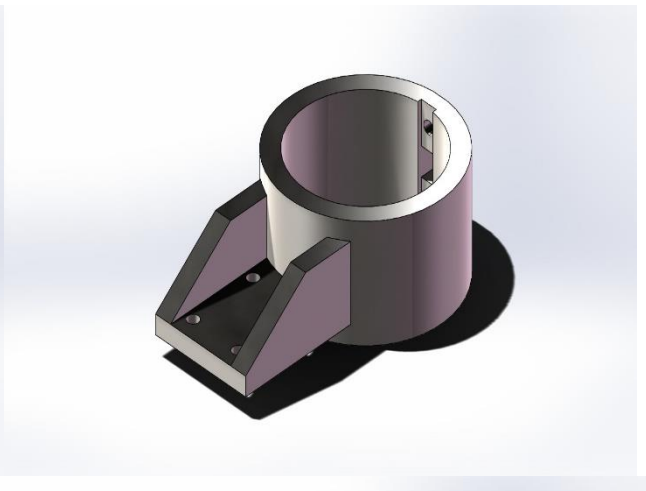
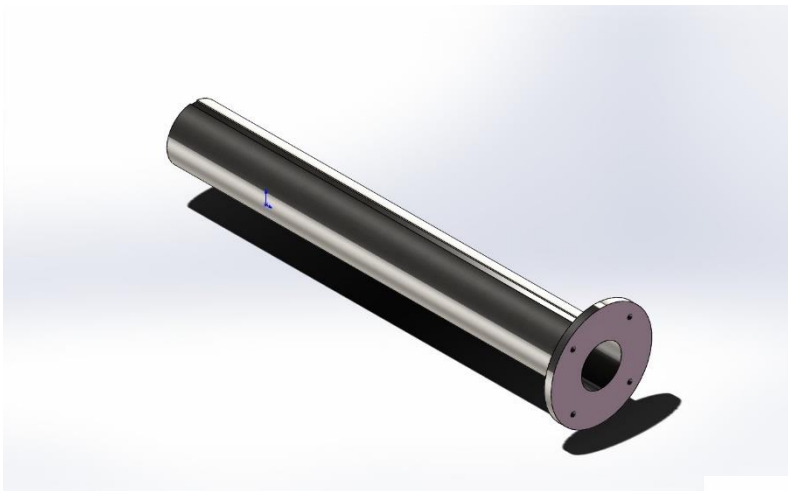


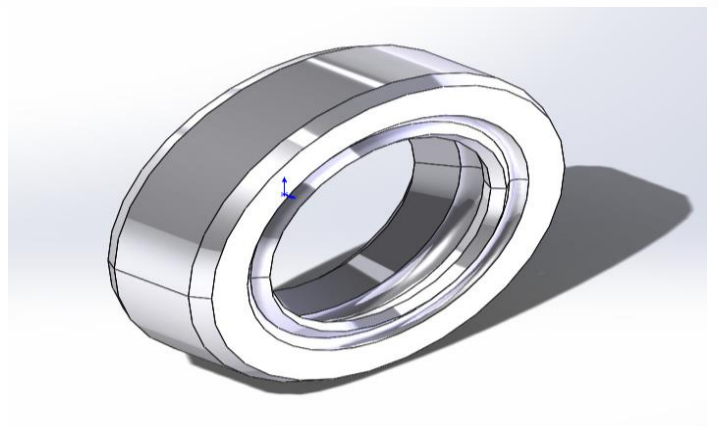
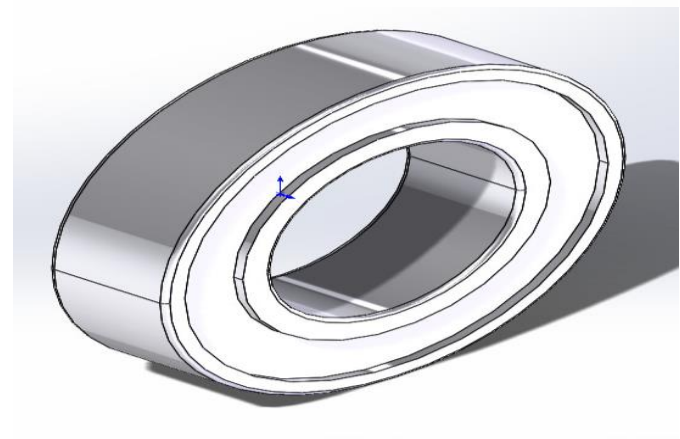
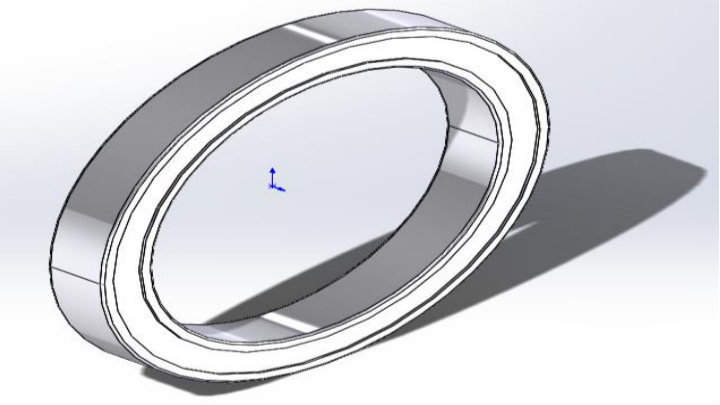
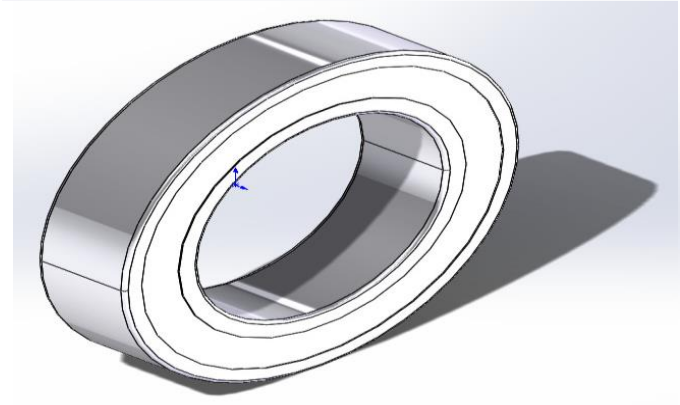
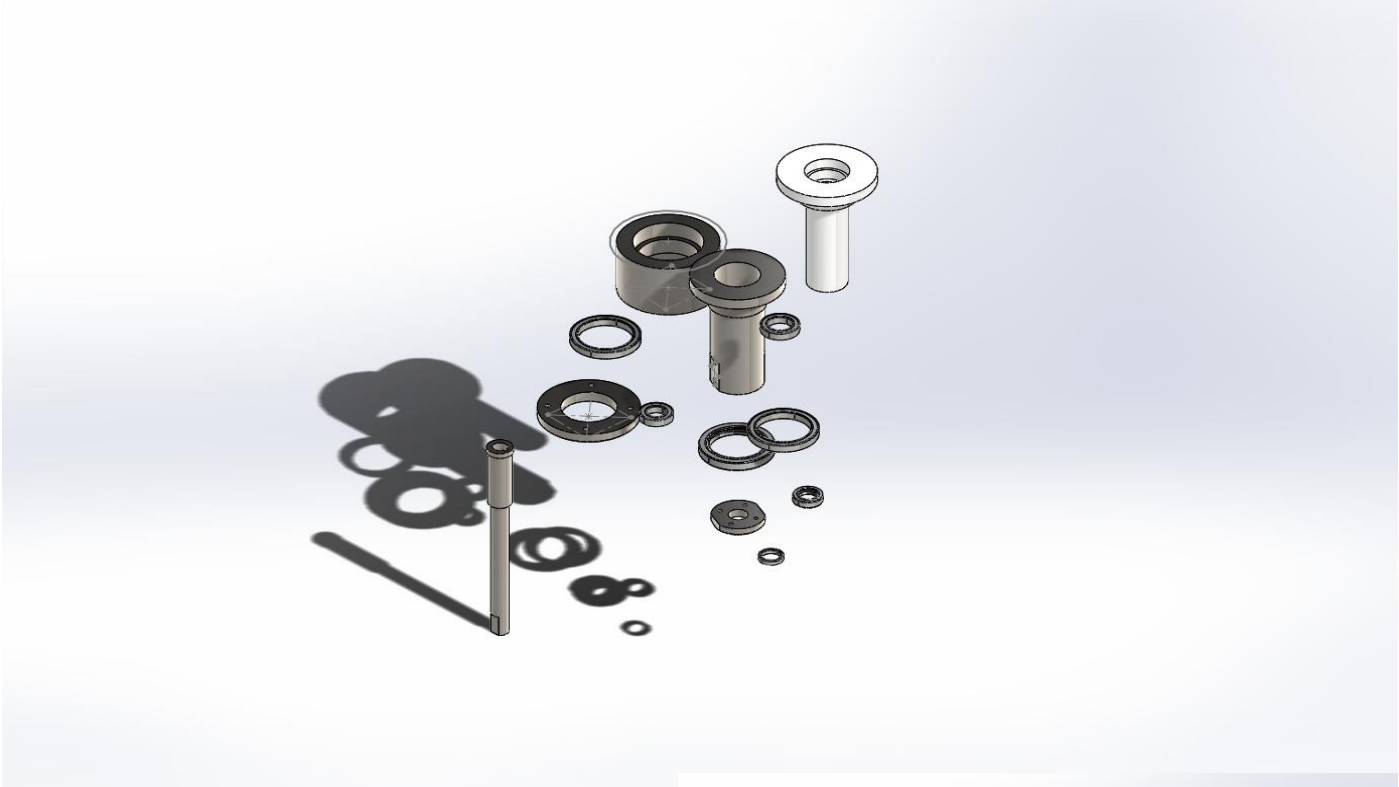
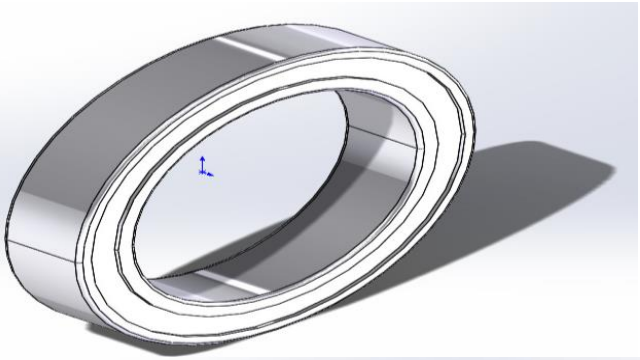
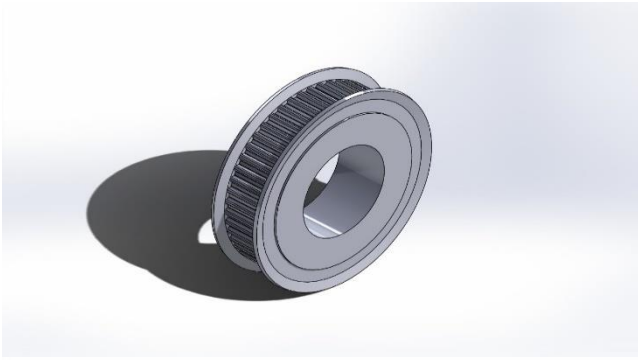
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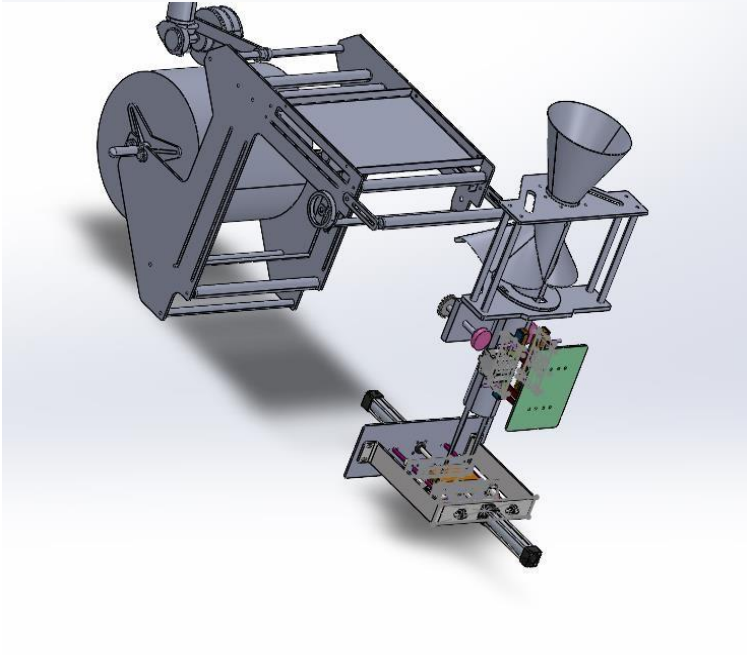
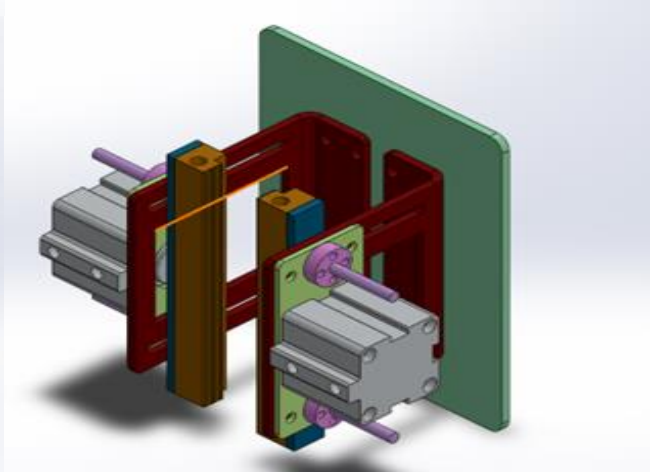
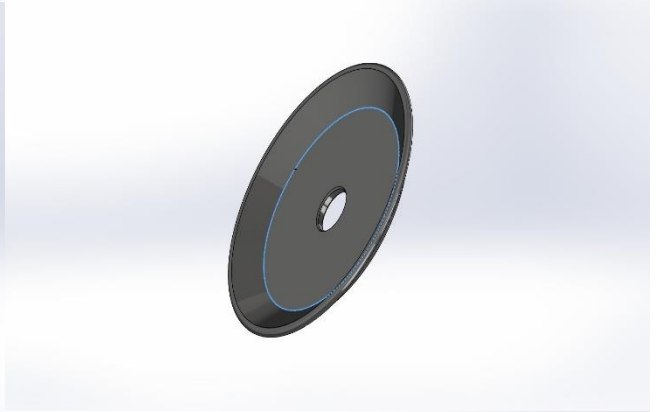
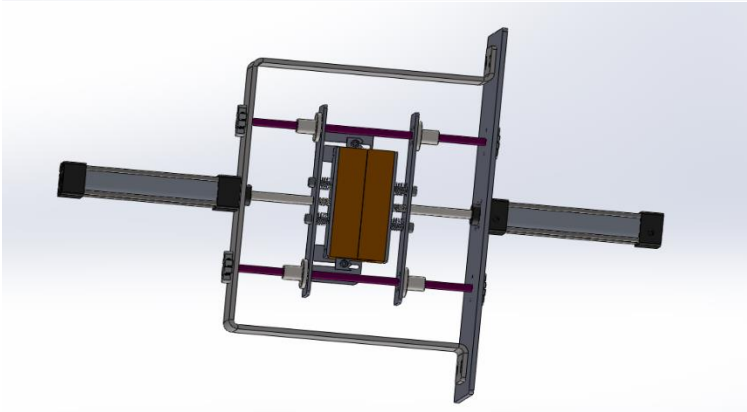
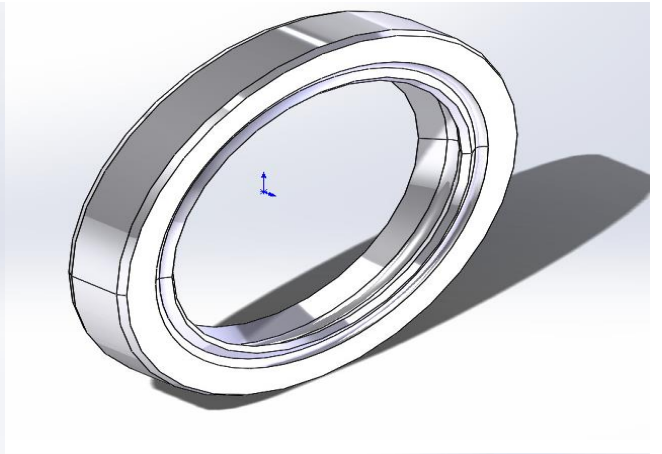
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A.4. Exploded View.

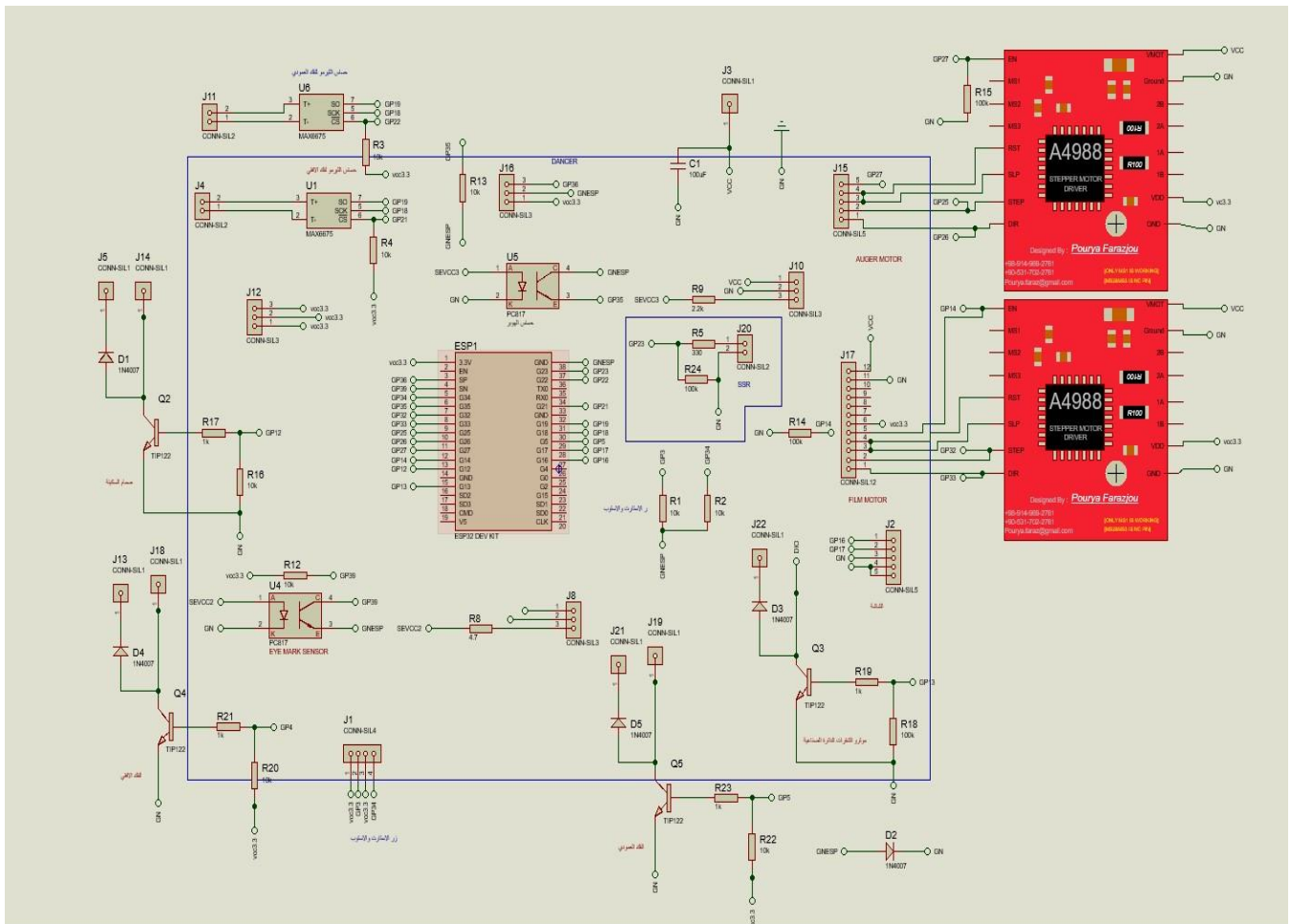




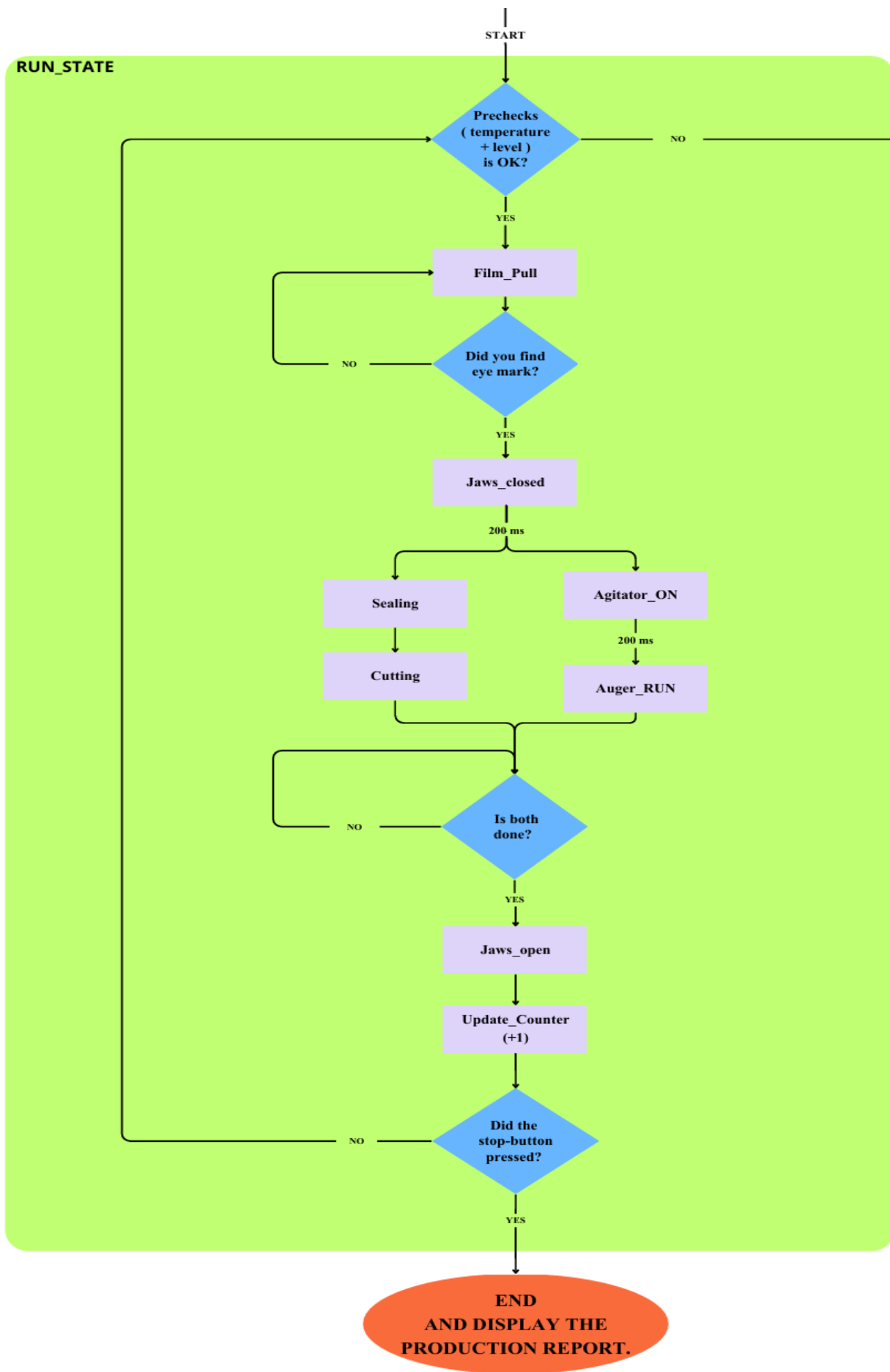




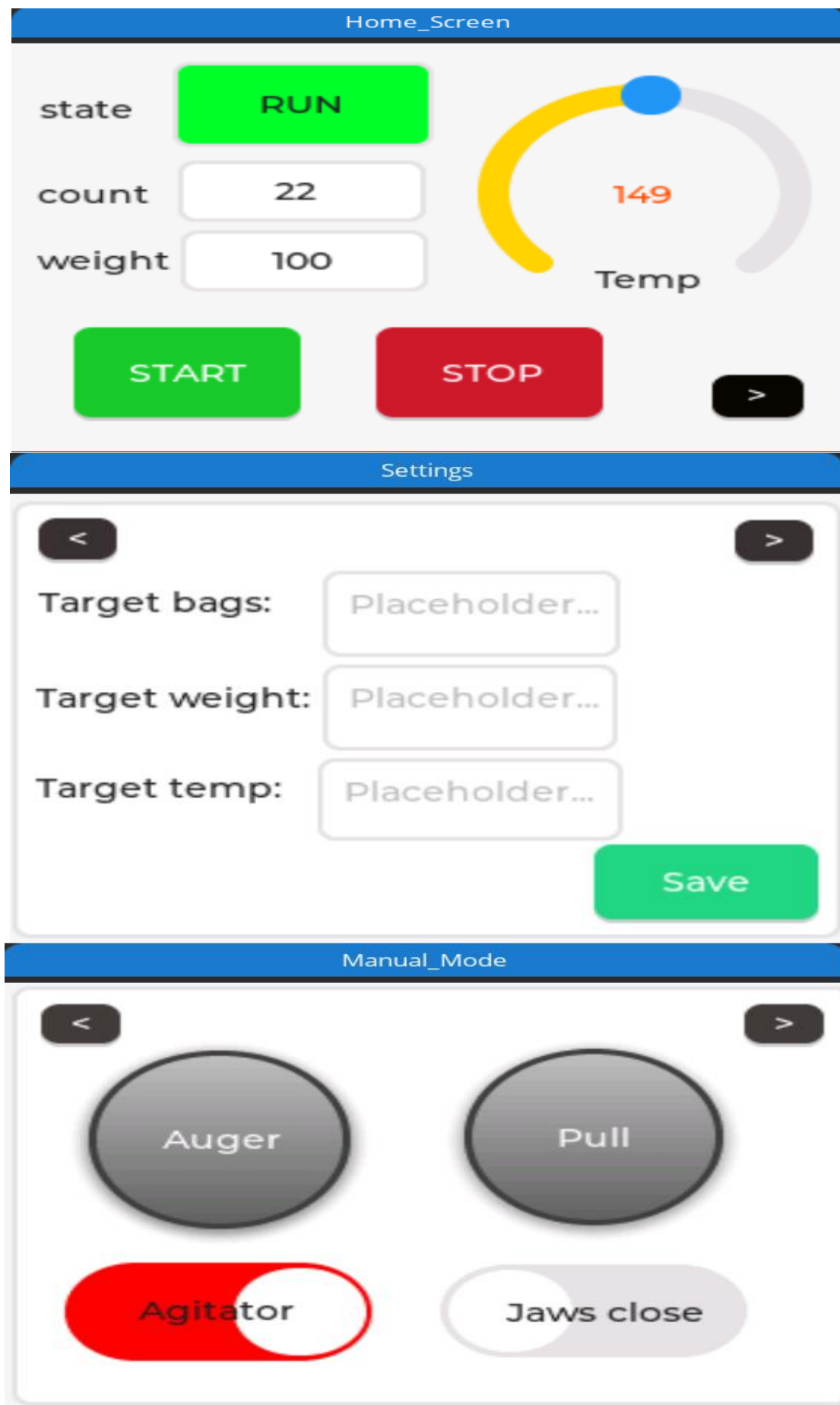
B.1. Wiring Diagram.



B.2. Control Logic Flowchart.



B.3. HMI Screen Mockups.



```

2 // VFFS Machine Controller - Graduation Project
3 // Features:
4 // 1. RMT Driver: Generates stepper pulses via Hardware (Non-blocking).
5 // 2. Hybrid Motion: RMT for speed (Blind Zone) + Manual for precision (Sensor Search).
6 // 3. Robust E-Stop: Kills RMT channels instantly via Fault Handler.
7 // 4. Optimized Core Usage: CPU 0 is now free for precise PID calculations.
8
9 #include <stdio.h>
10 #include <string.h>
11 #include <stdlib.h>
12 #include "freertos/FreeRTOS.h"
13 #include "freertos/task.h"
14 #include "freertos/queue.h"
15 #include "freertos/semphr.h"
16 #include "freertos/event_groups.h"
17 #include "driver/gpio.h"
18 #include "driver/rmt.h"
19 #include "esp_timer.h"
20 #include "esp_log.h"
21 #include "esp_rom_sys.h"
22 #include "esp_adc/adc_oneshot.h"
23 #include "mock_sensor.h"
24
25 // =====
26 // ***** USER CONFIGURATION AREA *****
27 // =====
28 #define SIMULATION_MODE 1
29 // PID & TEMP
30 #define HEATER_SETPPOINT 150.0
31 #define PID_KP 20.0
32 #define PID_KI 0.5
33 #define PID_KD 0.0
34 #define PID_WINDOW_SIZE_MS 1000
35 #define TEMP_TOLERANCE 5.0
36 // HOPPER
37 #define HOPPER_EMPTY_THRESH 10
38 #define HOPPER_LOW_THRESH 30
39 // BAG & MOTION
40 #define BAG_LENGTH_STEPS 700
41 #define BLIND_ZONE_STEPS 500
42 #define MAX_SEARCH_STEPS 1000
43 #define AUGER_STEPS 800
44 // SPEED (RMT tick is usually 1us)
45 #define SPEED_FILM_US 1000
46 #define SPEED_AUGER_US 1000
47 // DELAYS
48 #define DELAY_DJMS_CLOSE 200
49 #define DELAY_FILL 200
50 #define DELAY_CUT 200
51 #define DELAY_OPEN 200
52 #define DELAY_CYCLE_END 200
53 // =====
54 // PINS & DEFINITIONS
55 // =====
56 // --- RMT CHANNELS ---
57 #define RMT_CH_AUGER RMT_CHANNEL_0
58 #define RMT_CH_FILM RMT_CHANNEL_1
59 // --- SPI ---
60 #define SPI_PIN_SCK 18
61 #define SPI_PIN_MISO 19
62 #define SPI_PIN_CS_1 21
63 // --- GPIOs ---
64 #define PIN_ESTOP 15
65 #define PIN_START_BTN 34
66 #define PIN_STOP_BTN 2
67 #define PIN_EYE_MARK 39
68 // --- Motors ---
69 #define PIN_AUGER_STEP 25
70 #define PIN_AUGER_DIR 26
71 #define PIN_AUGER_EN 27
72 #define PIN_FILM_STEP 32
73 #define PIN_FILM_DIR 33
74 #define PIN_FILM_EN 14
75 // --- Actuators ---
76 #define PIN_AGITATOR 13
77 #define PIN_HEATER_SSR 23
78 #define PIN_VALVE_HORIZ 4
79 #define PIN_VALVE_VERT 5
80 #define PIN_VALVE_CUTTER 12
81
82 // --- ADC ---
83 #define ADC_UNIT_ID ADC_UNIT_1
84 #define ADC_CHAN_LEVEL ADC_CHANNEL_7
85
86 // --- Logic ---
87 #define VALVE_ON 0
88 #define VALVE_OFF 1
89 #define AGIT_ON 1
90 #define AGIT_OFF 0
91 #define MOTOR_ENABLE 0
92 #define MOTOR_DISABLE 1
93 #define MOTOR_AUGER_ID 1
94 #define MOTOR_FILM_ID 2
95
96 // --- Events ---
97 #define BIT_SYSTEM_READY (1 << 0)
98 #define BIT_AUTO_RUNNING (1 << 1)
99 #define BIT_ESTOP_ACTIVE (1 << 2)
100 #define BIT_TEMP_OK (1 << 3)
101 // =====
102 // HANDLES
103 // =====
104 // --- RMT ---
105 // =====
106 // --- HELPERS ---
107 // =====
108 void IRAM_ATTR low_gpio_handler(void* arg) {
109     BaseType_t higherPriorityTaskWoken = pdFALSE;
110     xEventGroupSetBitsFromISR(sys_evt_group, BIT_ESTOP_ACTIVE, &higherPriorityTaskWoken);
111     portYIELD_FROM_ISR(&higherPriorityTaskWoken);
112 }
113
114 int get_hopper_level() {
115     int raw_val = 0;
116     if (adc_oneshot_read(adc1_handle, ADC_CHAN_LEVEL, &raw_val) == ESP_OK) {
117         return (raw_val * 100) / 4095;
118     }
119     return 0;
120 }
121
122 void SendMotionCommand(uint8_t id, uint32_t steps, uint32_t speed_us, uint8_t dir, bool use_sensor) {
123     MotionCommand_t cmd = { .motor_id = id, .steps = steps, .interval_us = speed_us, .direction = dir, .use_sensor = use_sensor };
124     xQueueSend(motion_queue, &cmd, portMAX_DELAY);
125 }
126
127 bool WaitForMotionResult() {
128     while(!xSemaphoreTake(motion_done_semaphore, pdMS_TO_TICKS(100)) == pdFALSE) {
129         if (xEventGroupGetBits(sys_evt_group) & BIT_ESTOP_ACTIVE) return false;
130     }
131     bool success = true;
132     if (xSemaphoreTake(mutex_vars, pdMS_TO_TICKS(10))) {
133         success = sys_data.last_move_success;
134         xSemaphoreGive(mutex_vars);
135     }
136     return success;
137 }
138
139 // =====
140 // TASKS
141 // =====
142 // --- MOTION TASK (RMT ENABLED) ---
143 void Task_MotionController(void* pvParameters) {
144     MotionCommand_t cmd;
145     int dir_pin, en_pin;
146     rmt_channel_t rmt_chan;
147
148     for (;;) {
149         if (xQueueReceive(motion_queue, &cmd, portMAX_DELAY)) {
150             // 1. Setup
151             if (cmd.motor_id == MOTOR_AUGER_ID) {
152                 dir_pin = PIN_AUGER_DIR; en_pin = PIN_AUGER_EN; rmt_chan = RMT_CH_AUGER;
153             } else {
154                 dir_pin = PIN_FILM_DIR; en_pin = PIN_FILM_EN; rmt_chan = RMT_CH_FILM;
155             }
156             gpio_set_level(dir_pin, cmd.direction);
157             gpio_set_level(en_pin, MOTOR_ENABLE);
158             esp_rom_delay_us(50);
159
160             bool move_ok = true;
161
162             // 2. Execution Logic
163             if (cmd.use_sensor) {
164                 // --- HYBRID MODE: RMT for Blind Zone + Manual for Search ---
165                 printf("[NOTION] Hybrid Pull: RMT (Blind) -> Manual (Search)\n");
166
167                 // A) RMT Phase: Blind Zone (Fast & CPU Free)
168                 if (BLIND_ZONE_STEPS > 0) {
169                     // Create RMT items buffer
170                     size_t size = BLIND_ZONE_STEPS * sizeof(rmt_item32_t);
171                     rmt_item32_t* items = (rmt_item32_t*) malloc(size);
172
173                     if (items) {
174                         for (int i = 0; i < BLIND_ZONE_STEPS; i++) {
175                             // High/Low duration = half interval
176

```

```

227 // rmt_items_t items = (rmt_items_t){0};
228
229 if (items) {
230     for (int i = 0; i < BLIND_ZONE_STEPS; i++) {
231         // High/Low duration = half interval
232         items[i] = (rmt_item32_t){(cmd.interval_us/2, 1, cmd.interval_us/2, 0)};
233     }
234     // Send and Wait
235     rmt_write_items(rmt_chan, items, BLIND_ZONE_STEPS, true);
236     rmt_wait_tx_done(rmt_chan, portMAX_DELAY); // Wait for HW to finish
237     free(items);
238 }
239
240 // Check Safety after RMT
241 if (xEventGroupGetBits(sys_evt_group) & BIT_ESTOP_ACTIVE) move_ok = false;
242
243 // B) Manual Phase: Search Zone (Precise)
244 if (move_ok) {
245     bool mark_found = false;
246     int step_pin = (cmd.motor_id == MOTOR_AUGER_ID) ? PIN_AUGER_STEP : PIN_FILM_STEP;
247
248     uint32_t remaining_search = MAX_SEARCH_STEPS - BLIND_ZONE_STEPS;
249     for (uint32_t i = 0; i < remaining_search; i++) {
250         if (xEventGroupGetBits(sys_evt_group) & BIT_ESTOP_ACTIVE) { move_ok = false; break; }
251
252         gpio_set_level(step_pin, 1); esp_rom_delay_us(cmd.interval_us / 2);
253         gpio_set_level(step_pin, 0); esp_rom_delay_us(cmd.interval_us / 2);
254
255         if (gpio_get_level(PIN_EYE_MARK) == 0) { // Found!
256             mark_found = true;
257             printf("[MOTION] Mark Found in Search Zone.\n");
258             break;
259         }
260     }
261     if (!mark_found && move_ok) {
262         printf("[MOTION] Error: Mark Missed!\n");
263         move_ok = false;
264     }
265 }
266
267 } else {
268     // === PURE RMT MODE (Auger) ===
269     // Generate all pulses via hardware
270     size_t size = cmd.steps * sizeof(rmt_item32_t);
271     rmt_item32_t* items = (rmt_item32_t*) malloc(size);
272
273     if (items) {
274         for (int i = 0; i < cmd.steps; i++) {
275             items[i] = (rmt_item32_t){(cmd.interval_us/2, 1, cmd.interval_us/2, 0)};
276         }

```

```

320     printf("[PID] T.K.If C | Pwr:%.0f%%\n", temp_current, (pid_output/PID_WINDOW_SIZE_MS)*100.0);
321     error = HEATER_SETPPOINT - temp_current;
322     integral += error;
323     if (integral > 100.0) integral = 100.0;
324     if (integral < -100.0) integral = -100.0;
325
326     double output_val = (PID_KP * error) + (PID_KI * integral) + (PID_KD * (error - last_error));
327     last_error = error;
328
329     if (output_val > PID_WINDOW_SIZE_MS) output_val = PID_WINDOW_SIZE_MS;
330     if (output_val < 0) output_val = 0;
331     pid_output = output_val;
332
333     if (temp_current > (HEATER_SETPPOINT - TEMP_TOLERANCE)) xEventGroupSetBits(sys_evt_group, BIT_TEMP_OK
334     else xEventGroupClearBits(sys_evt_group, BIT_TEMP_OK);
335 }
336
337 if (in_window < pid_output) gpio_set_level(PIN_HEATER_SSR, 1);
338 else gpio_set_level(PIN_HEATER_SSR, 0);
339
340 vTaskDelay(pdMS_TO_TICKS(20));
341 }
342
343 // --- LOGIC & HMI TASKS (UNCHANGED) ---
344 void Task_Controller(void *pvParameters) {
345     vTaskDelay(pdMS_TO_TICKS(1000));
346     xEventGroupSetBits(sys_evt_group, BIT_SYSTEM_READY);
347
348     for (;;) {
349         EventBits_t bits = xEventGroupWaitBits(sys_evt_group, BIT_AUTO_RUNNING, pdFALSE, pdTRUE, pdMS_TO_TICKS(
350
351         if (bits & BIT_AUTO_RUNNING) {
352             if (!!(xEventGroupGetBits(sys_evt_group) & BIT_TEMP_OK)) {
353                 printf("[SYS] Paused: Waiting for Temp...\n");
354                 vTaskDelay(pdMS_TO_TICKS(1000)); continue;
355             }
356         }
357
358         int level = get_hopper_level();
359         if (level < HOPPER_EMPTY_THRESH) {
360             printf("[ALARM] Hopper Empty!\n");
361             xEventGroupClearBits(sys_evt_group, BIT_AUTO_RUNNING); continue;
362         } else if (level < HOPPER_LOW_THRESH) {
363             printf("[WARNING] Low Material (%d%%).\n", level);
364         }
365
366         printf("[CYCLE] 1. Pulling Film...\n");
367         SendMotionCommand(MOTOR_FILM_ID, 0, SPEED_FILM_US, 1, true);
368         if (!WaitForMotionResult()) {
369             printf("[FAULT] Film Error.\n");
370             xEventGroupClearBits(sys_evt_group, BIT_AUTO_RUNNING); continue;

```

```

274     for (int i = 0; i < cmd.steps; i++) {
275         items[i] = (rmt_item32_t){(cmd.interval_us/2, 1, cmd.interval_us/2, 0)};
276     }
277     rmt_write_items(rmt_chan, items, cmd.steps, true);
278
279     // While RMT works, we wait. If E-Stop happens, FaultHandler kills RMT.
280     // We just need to wait for completion or abort.
281     while (rmt_wait_tx_done(rmt_chan, 10 * portTICK_PERIOD_MS) != ESP_OK) {
282         if (xEventGroupGetBits(sys_evt_group) & BIT_ESTOP_ACTIVE) {
283             move_ok = false; break;
284         }
285     }
286     free(items);
287 }
288
289 gpio_set_level(en_pin, MOTOR_DISABLE);
290 if (xSemaphoreTake(mutex_vars, pdMS_TO_TICKS(10))) {
291     sys_data.last_move_success = move_ok;
292     xSemaphoreGive(mutex_vars);
293 }
294 xSemaphoreGive(motion_done_sem);
295 }
296
297 }
298
299 // --- PID TASK (Precise due to RMT) ---
300 void Task_PID(void *pvParameters) {
301     // Setup SPI
302     gpio_set_direction(SPI_PIN_CS_1, GPIO_MODE_OUTPUT);
303     gpio_set_direction(SPI_PIN_SCK, GPIO_MODE_OUTPUT);
304     gpio_set_direction(SPI_PIN_MISO, GPIO_MODE_INPUT);
305     gpio_set_level(SPI_PIN_CS_1, 1);
306     double temp_current = 0.0, pid_output = 0.0;
307     double error = 0.0, last_error = 0.0, integral = 0.0;
308     int64_t window_start = esp_timer_get_time() / 1000;
309     for (;;) {
310         int64_t now_ms = esp_timer_get_time() / 1000;
311         int64_t in_window = (now_ms - window_start);
312
313         if (in_window >= PID_WINDOW_SIZE_MS) {
314             window_start = now_ms;
315             in_window = 0;
316             temp_current = Read_Temperature_Sensor();
317             if (temp_current == -1.0) temp_current = sys_data.current_temp; // fallback
318             if (xSemaphoreTake(mutex_vars, pdMS_TO_TICKS(10))) {
319                 sys_data.current_temp = temp_current;
320                 xSemaphoreGive(mutex_vars);
321             }
322             printf("[PID] T.K.If C | Pwr:%.0f%%\n", temp_current, (pid_output/PID_WINDOW_SIZE_MS)*100.0);
323             error = HEATER_SETPPOINT - temp_current;
324             integral += error;
325             if (integral > 100.0) integral = 100.0;

```

```

371         xEventGroupClearBits(sys_evt_group, BIT_AUTO_RUNNING); continue;
372     }
373
374     printf("[CYCLE] 2. Jaws Closing...\n");
375     gpio_set_level(PIN_VALVE_HORIZ, VALVE_ON);
376     gpio_set_level(PIN_VALVE_VERT, VALVE_ON);
377     vTaskDelay(pdMS_TO_TICKS(Delay_Jaws_Close));
378
379     printf("[CYCLE] 3. Filling...\n");
380     gpio_set_level(PIN_AGITATOR, AGIT_ON);
381     vTaskDelay(pdMS_TO_TICKS(Delay_Fill));
382     SendMotionCommand(MOTOR_AUGER_ID, AUGER_STEPS, SPEED_AUGER_US, 1, false);
383     WaitForMotionResult();
384
385     printf("[CYCLE] >> CUT!\n");
386     gpio_set_level(PIN_VALVE_CUTTER, VALVE_ON);
387     vTaskDelay(pdMS_TO_TICKS(Delay_Cut));
388     gpio_set_level(PIN_VALVE_CUTTER, VALVE_OFF);
389
390     printf("[CYCLE] 4. Open Jaws!\n");
391     gpio_set_level(PIN_VALVE_VERT, VALVE_OFF);
392     gpio_set_level(PIN_AGITATOR, AGIT_OFF);
393     printf("[CYCLE] 6. Release Horizontal!\n");
394     gpio_set_level(PIN_VALVE_HORIZ, VALVE_OFF);
395     vTaskDelay(pdMS_TO_TICKS(Delay_Open));
396
397     if (xSemaphoreTake(mutex_vars, pdMS_TO_TICKS(10))) {
398         sys_data.bag_counter++;
399         printf("[DATA] Bags: %d\n", sys_data.bag_counter);
400         xSemaphoreGive(mutex_vars);
401     }
402     vTaskDelay(pdMS_TO_TICKS(Delay_Cycle_End));
403
404     } else {
405         vTaskDelay(pdMS_TO_TICKS(500));
406     }
407 }
408
409 void Task_HMI(void *pvParameters) {
410     int last_start = 0, last_stop = 0;
411     for (;;) {
412         int start = gpio_get_level(PIN_START_BTN);
413         if (start == 1 && last_start == 0) {
414             EventBits_t status = xEventGroupGetBits(sys_evt_group);
415             if (status & BIT_ESTOP_ACTIVE) {
416                 printf("[HMI] REJECTED: E-STOP Active!\n");
417             } else if ((status & BIT_TEMP_OK) == 0) {
418                 printf("[HMI] REJECTED: Warming Up... (Wait for %.1f C)\n", HEATER_SETPPOINT);
419             } else {
420                 printf("[HMI] Start Accepted!\n");
421                 xEventGroupSetBits(sys_evt_group, BIT_AUTO_RUNNING);
422             }

```

```

412 int start = gpio_get_level(PIN_START_BTN);
413 if (start == 1 && last_start == 0) {
414     EventBits_t status = xEventGroupGetBits(sys_evt_group);
415     if (status & BIT_ESTOP_ACTIVE) {
416         printf("[HMI] REJECTED: E-STOP Active!\n");
417     } else if ((status & BIT_TEMP_OK) == 0) {
418         printf("[HMI] REJECTED: Warming Up... (Wait for %i C)\n", HEATER_SETPPOINT);
419     } else {
420         printf("[HMI] Start Accepted!\n");
421         xEventGroupSetBits(sys_evt_group, BIT_AUTO_RUNNING);
422     }
423 }
424 last_start = start;
425 int stop = gpio_get_level(PIN_STOP_BTN);
426 if (stop == 1 && last_stop == 0) {
427     printf("[HMI] Stop Received!\n");
428     xEventGroupClearBits(sys_evt_group, BIT_AUTO_RUNNING);
429 }
430 last_stop = stop;
431 vTaskDelay(pdMS_TO_TICKS(50));
432 }
433 }
434 // --- FAULT HANDLER (CRITICAL: RMT STOP) ---
435 void Task_FaultHandler(void *pvParameters) {
436     EventBits_t bits;
437     for (;;) {
438         bits = xEventGroupWaitBits(sys_evt_group, BIT_ESTOP_ACTIVE, pdFALSE, pdFALSE, portMAX_DELAY);
439         if (bits & BIT_ESTOP_ACTIVE) {
440             printf("\n!!! E-STOP TRIGGERED !!!\n");
441
442             // 1. Kill RMT Immediately (Hardware Stop)
443             rmt_tx_stop(RMT_CH_AUGER);
444             rmt_tx_stop(RMT_CH_FILM);
445
446             // 2. Suspend Tasks
447             if (th_ControlLoop) vTaskSuspend(th_ControlLoop);
448             if (th_PID) vTaskSuspend(th_PID);
449             if (th_Motion) vTaskSuspend(th_Motion);
450
451             // 3. Kill Outputs
452             gpio_set_level(PIN_AUGER_EN, MOTOR_DISABLE);
453             gpio_set_level(PIN_FILM_EN, MOTOR_DISABLE);
454             gpio_set_level(PIN_HEATER_SSR, 0);
455             gpio_set_level(PIN_VALVE_HORIZ, VALVE_OFF);
456             gpio_set_level(PIN_VALVE_VERT, VALVE_OFF);
457             gpio_set_level(PIN_VALVE_CUTTER, VALVE_OFF);
458             gpio_set_level(PIN_AGITATOR, AGIT_OFF);
459             xEventGroupClearBits(sys_evt_group, BIT_AUTO_RUNNING);
460
461             while(gpio_get_level(PIN_ESTOP) == 0) vTaskDelay(pdMS_TO_TICKS(50));
462             printf("[SYS] E-STOP Released. Resetting...\n");
463         }
464     }
465 }

```

```

509
510 // Buttons
511 io_conf.mode = GPIO_MODE_INPUT;
512 io_conf.pin_bit_mask = (1ULL<<PIN_START_BTN) | (1ULL<<PIN_STOP_BTN);
513 io_conf.pull_up_en = 0; io_conf.pull_down_en = 0;
514 gpio_config(&io_conf);
515
516 // Sensors
517 io_conf.pin_bit_mask = (1ULL<<PIN_EYE_MARK) | (1ULL<<SPI_PIN_MISO);
518 io_conf.pull_up_en = 1; io_conf.pull_down_en = 0;
519 gpio_config(&io_conf);
520
521 // E-Stop
522 io_conf.intr_type = GPIO_INTR_NEGEDGE;
523 io_conf.pin_bit_mask = (1ULL<<PIN_ESTOP);
524 io_conf.pull_up_en = 0;
525 gpio_config(&io_conf);
526
527 gpio_install_isr_service(0);
528 gpio_isr_handler_add(PIN_ESTOP, isr_estop_handler, NULL);
529
530 //
531 Init_RMT(); // مهيئة نقطة RMT
532
533 #if SIMULATION_MODE
534     Start_Mock_Engine();
535 #endif
536
537 // Core Distribution
538 xTaskCreatePinnedToCore(Task_FaultHandler, "FaultTask", 2048, NULL, 5, NULL, 0);
539 xTaskCreatePinnedToCore(Task_MotionController, "MotionTask", 4096, NULL, 4, &th_Motion, 0);
540 xTaskCreatePinnedToCore(Task_PID, "PIDTask", 2048, NULL, 3, &th_PID, 0);
541
542 xTaskCreatePinnedToCore(Task_ControlLoop, "ControlTask", 4096, NULL, 2, &th_ControlLoop, 1);
543 xTaskCreatePinnedToCore(Task_HMI, "HMITask", 2048, NULL, 1, &th_HMI, 1);
544
545 printf("System Ready (RMT Accelerated).\n");
546 }

```

```

464 // Respawn
465 if (th_Motion) { vTaskDelete(th_Motion); th_Motion = NULL; }
466 xTaskCreatePinnedToCore(Task_MotionController, "MotionTask", 4096, NULL, 5, &th_Motion, 0);
467 if (th_ControlLoop) { vTaskDelete(th_ControlLoop); th_ControlLoop = NULL; }
468 xTaskCreatePinnedToCore(Task_ControlLoop, "ControlTask", 4096, NULL, 2, &th_ControlLoop, 1);
469 if (th_PID) { vTaskDelete(th_PID); th_PID = NULL; }
470 xTaskCreatePinnedToCore(Task_PID, "PIDTask", 2048, NULL, 3, &th_PID, 0);
471
472 xEventGroupClearBits(sys_evt_group, BIT_ESTOP_ACTIVE);
473 }
474 }
475 }
476
477 void app_main() {
478     sys_evt_group = xEventGroupCreate();
479     mutex_vars = xSemaphoreCreateMutex();
480     motion_queue = xQueueCreate(5, sizeof(MotionCommand_t));
481     motion_done_sem = xSemaphoreCreateBinary();
482
483     adc_oneshot_unit_init_cfg_t init_cfg1 = { .unit_id = ADC_UNIT_ID };
484     adc_oneshot_new_unit(&init_cfg1, &adc1_handle);
485     adc_oneshot_chan_cfg_t config = { .bitwidth = ADC_BITWIDTH_DEFAULT, .atten = ADC_ATTEN_DB_12 };
486     adc_oneshot_config_channel(&adc1_handle, ADC_CHAN_LEVEL, &config);
487
488     gpio_config_t io_conf = {};
489     io_conf.intr_type = GPIO_INTR_DISABLE;
490
491     // Heater
492     io_conf.mode = GPIO_MODE_INPUT_OUTPUT;
493     io_conf.pin_bit_mask = (1ULL<<PIN_HEATER_SSR);
494     io_conf.pull_down_en = 0; io_conf.pull_up_en = 0;
495     gpio_config(&io_conf);
496
497     // Outputs
498     io_conf.mode = GPIO_MODE_OUTPUT;
499     gpio_set_level(PIN_VALVE_HORIZ, VALVE_OFF);
500     gpio_set_level(PIN_VALVE_VERT, VALVE_OFF);
501     gpio_set_level(PIN_VALVE_CUTTER, VALVE_OFF);
502     gpio_set_level(PIN_AGITATOR, AGIT_OFF);
503     io_conf.pin_bit_mask = (1ULL<<PIN_AUGER_EN) | (1ULL<<PIN_AUGER_DIR) | // No STEP here (RMT handles it)
504         (1ULL<<PIN_FILM_EN) | (1ULL<<PIN_FILM_DIR) |
505         (1ULL<<PIN_AGITATOR) |
506         (1ULL<<PIN_VALVE_HORIZ) | (1ULL<<PIN_VALVE_VERT) | (1ULL<<PIN_VALVE_CUTTER) |
507         (1ULL<<SPI_PIN_CS_1) | (1ULL<<SPI_PIN_SCK);
508     gpio_config(&io_conf);
509
510     // Buttons
511     io_conf.mode = GPIO_MODE_INPUT;
512     io_conf.pin_bit_mask = (1ULL<<PIN_START_BTN) | (1ULL<<PIN_STOP_BTN);
513     io_conf.pull_up_en = 0; io_conf.pull_down_en = 0;
514     gpio_config(&io_conf);
515 }

```



Appendix C: Simulation and Data Reports

C.1. Mesh Sensitivity Analysis Data.

Table C.1 presents how results converged with different mesh sizes.

Platform	Mesh Type	Element Size	Elements	Max Stress (MPa)
SOLIDWORKS	Coarse	-	19779	21.3
SOLIDWORKS	Medium	-	50377	85.37
SOLIDWORKS	Fine	-	351350	127.3
ANSYS	Coarse	10 mm	52340	62.1
ANSYS	Medium	5 mm	118450	85.92
ANSYS	Fine	2.5 mm	554032	86.39

C.2. Material Datasheets: Technical properties of the powder used (e.g., Flour, Milk Powder) and the Stainless Steel grade.

Table C.2: Bulk Density and Flow Properties of Target Powders

Material	Bulk Density (kg/m ³)	AoR (°)	Cohesion (Pa)	Flow Function
Coffee Powder	250-400	30-40	150-300	Cohesive
Granulated Sugar	700-880	30-36	50-100	Easy-flowing
Wheat Flour	560-600	38-45	300-500	Very cohesive
Fine Table Salt	1200-1400	32-35	100-200	Easy-flowing
Powdered Detergent	300-600	30-45	200-400	Cohesive
Milk Powder	200-720	35-44	250-450	Cohesive

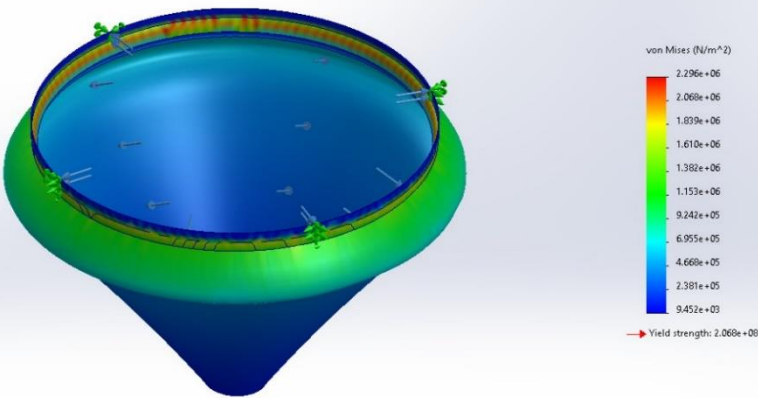
Material Properties

Properties	
Name:	AISI 304
Model type:	Linear Elastic Isotropic
Default failure criterion:	Max von Mises Stress
Yield strength:	2.06807e+08 N/m ²
Tensile strength:	5.17017e+08 N/m ²
Elastic modulus:	1.9e+11 N/m ²
Poisson's ratio:	0.29
Mass density:	8,000 kg/m ³
Shear modulus:	7.5e+10 N/m ²
Thermal expansion coefficient:	1.8e-05 /Kelvin



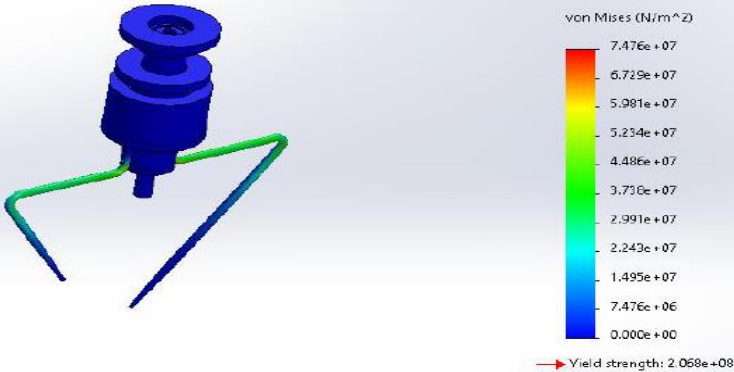
C.3. Extended Simulation Plots.

Model name: hopper2
Study name: Static 1 (-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 71,489.8

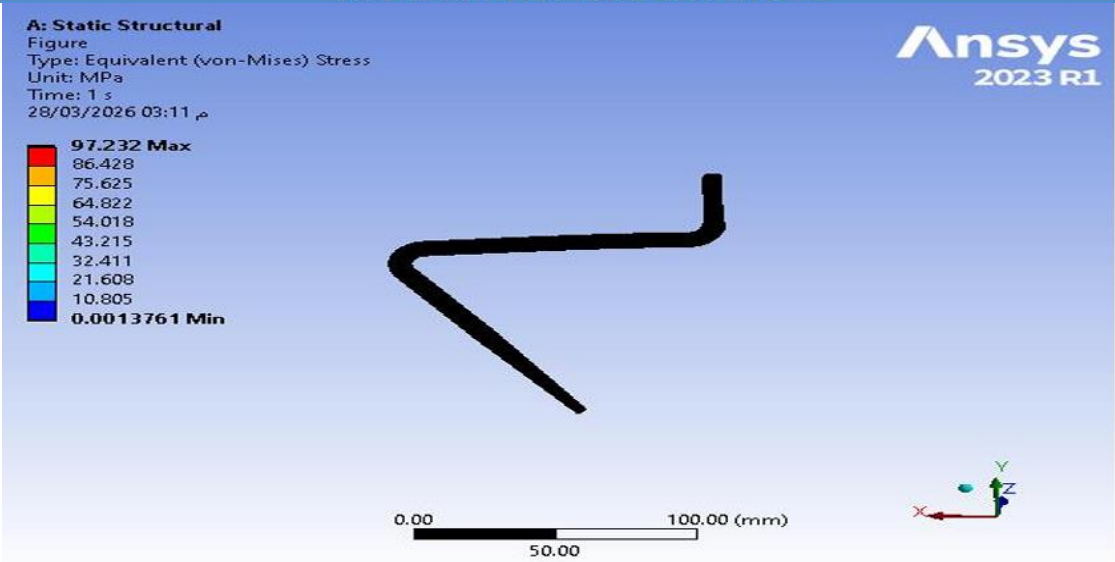


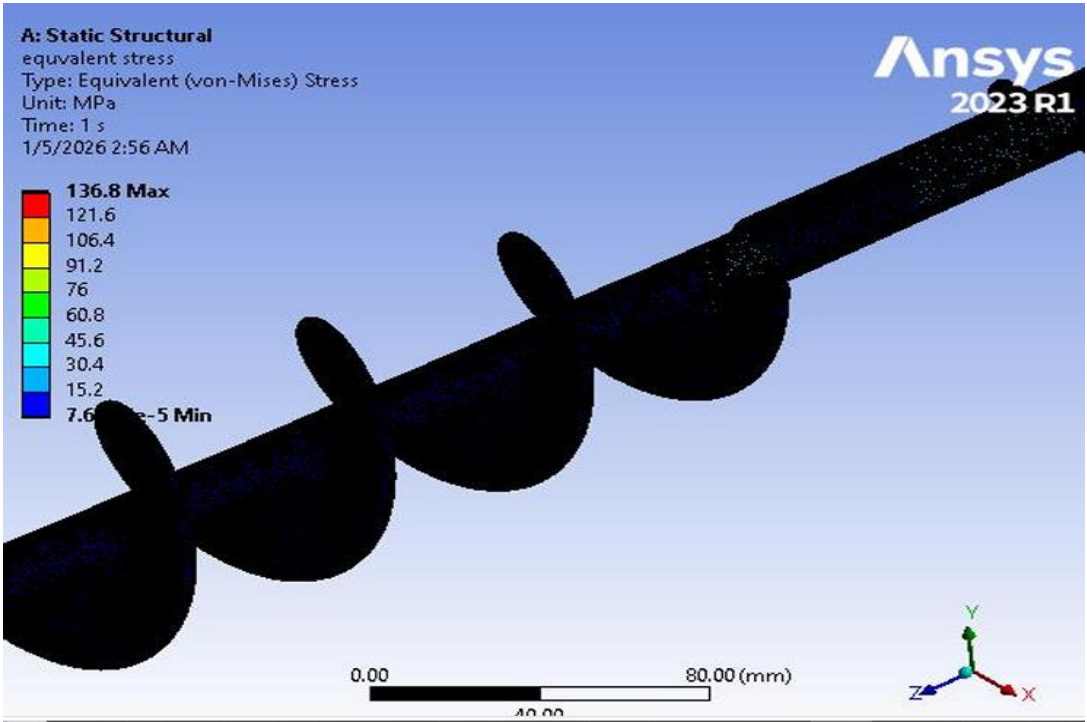
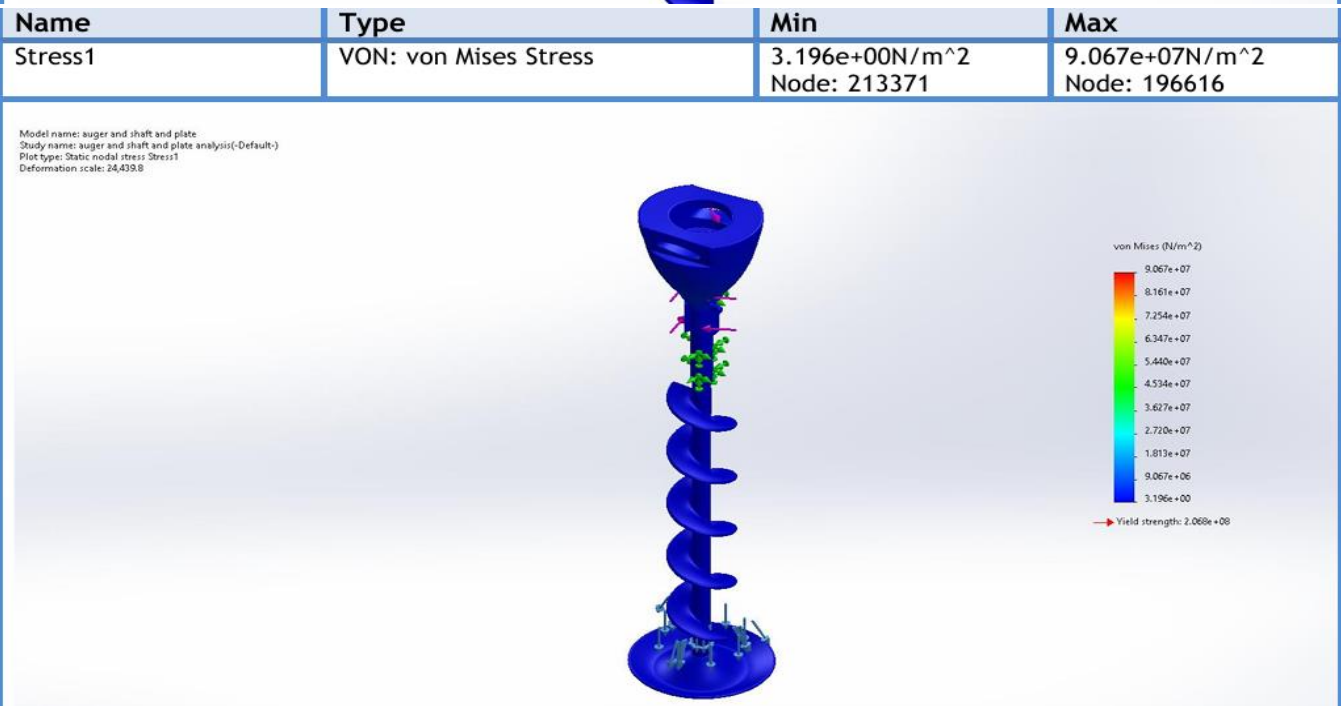
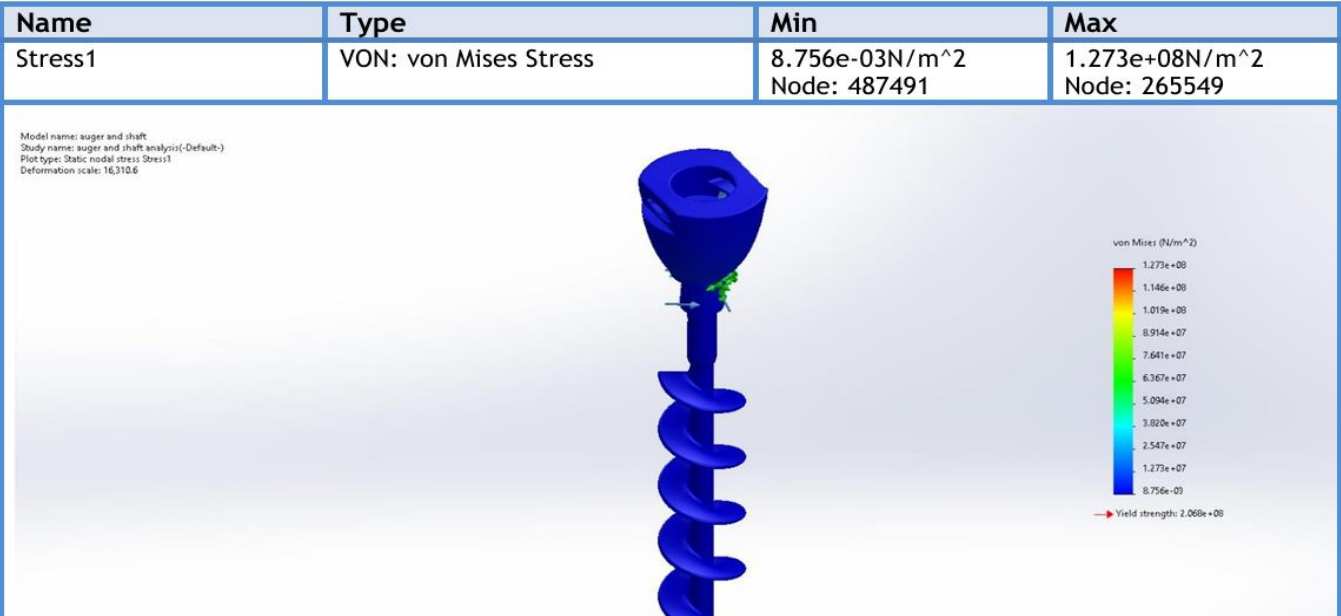
Name	Type	Min	Max
Stress1	VON: von Mises Stress	0.000e+00N/m^2 Node: 70847	7.476e+07N/m^2 Node: 437692

Model name: Assem4
Study name: assembly_1_static (-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 193418

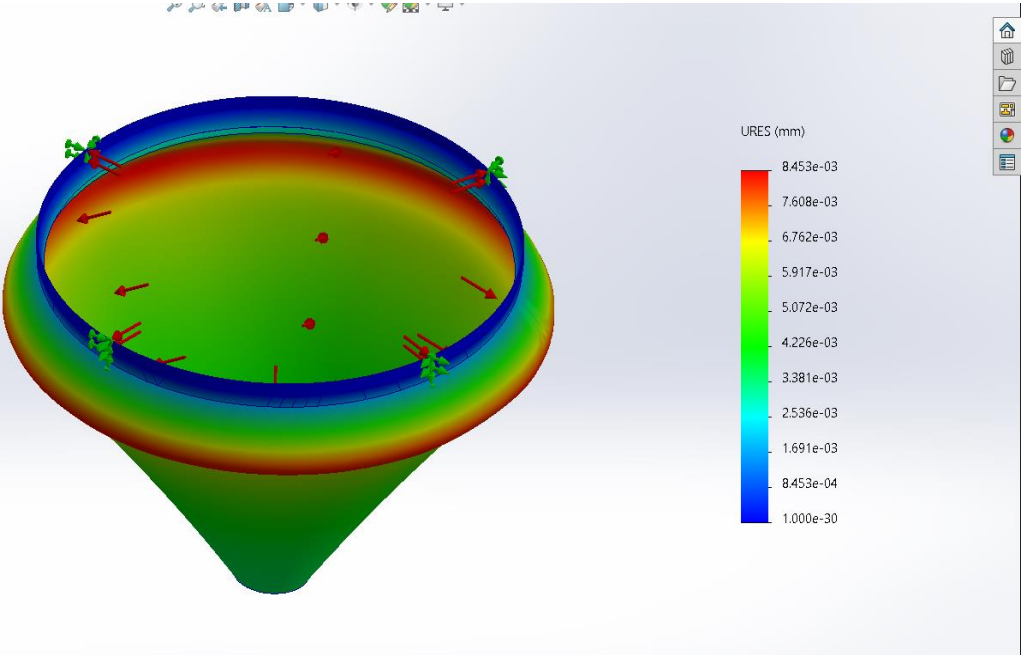


Assem4-assembly_1_static-Stress-Stress1



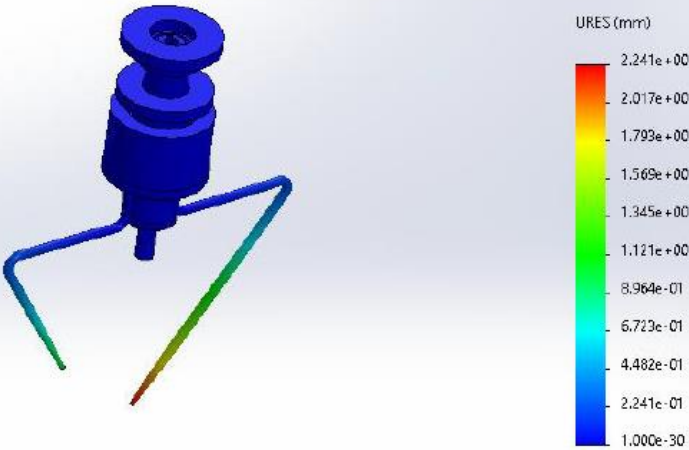


Model name: hopper2
Study name: Static 1 (-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 7,683.5

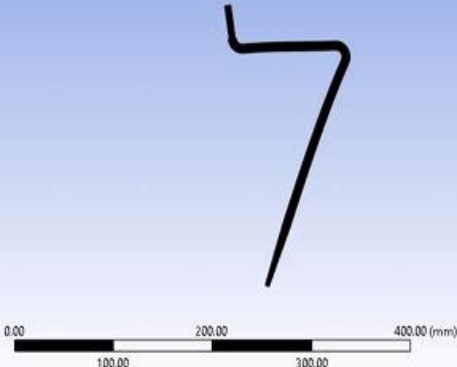
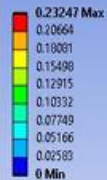


Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00mm Node: 70847	2.241e+00mm Node: 426121

Model name: Assem4
Study name: assembly_1_static(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 19.9418



A: Static Structural
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1 s
2/24/2026 10:14 PM



A: Static Structural

Figure

Type: Maximum Shear Stress

Unit: MPa

Time: 1 s

28/03/2026 03:11

Ansys
2023 R1

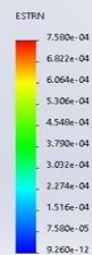
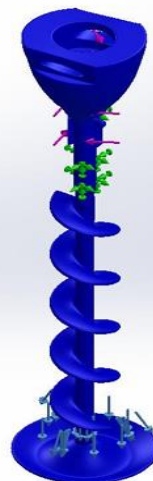


0.00 100.00 (mm)
50.00



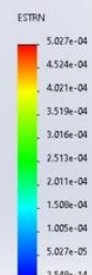
Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	9.260e-12 Element: 101310	7.580e-04 Element: 82398

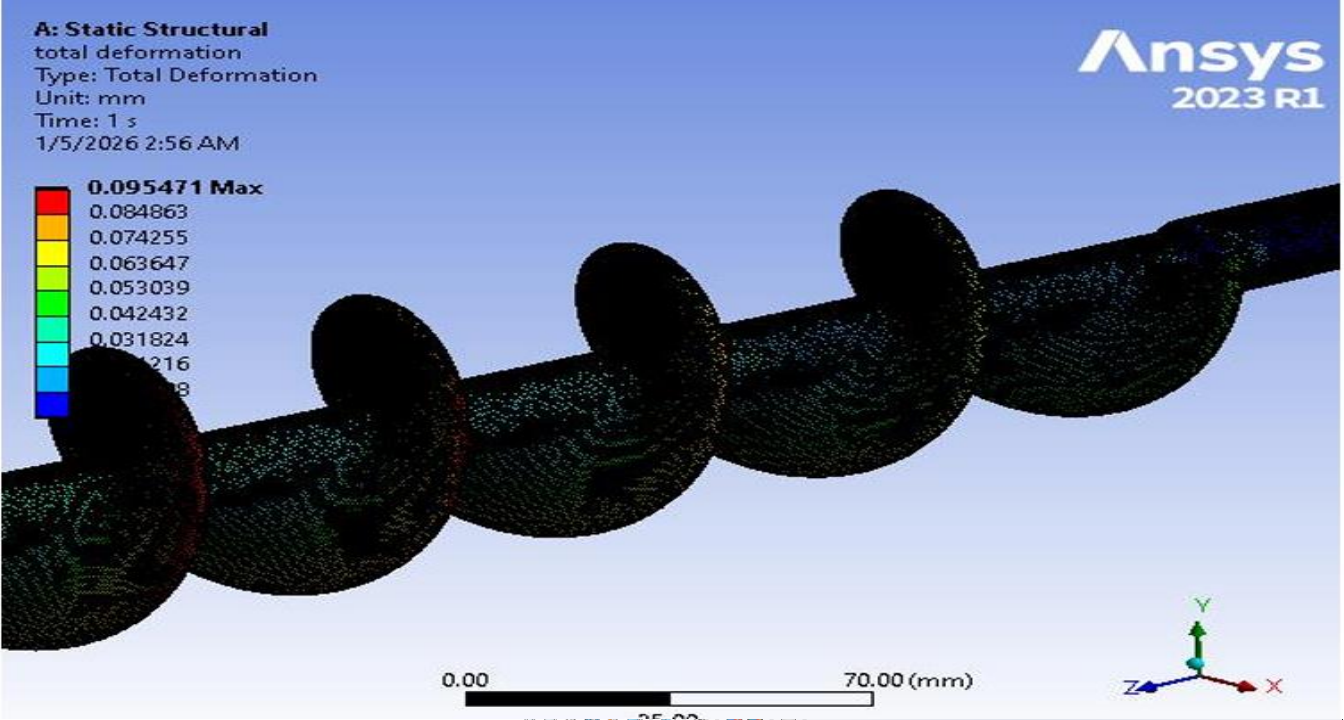
Model name: auger and shaft and plate
Study name: auger and shaft and plate analysis(-Default-)
Plot type: Static strain Strain1
Deformation scale: 24,439.8



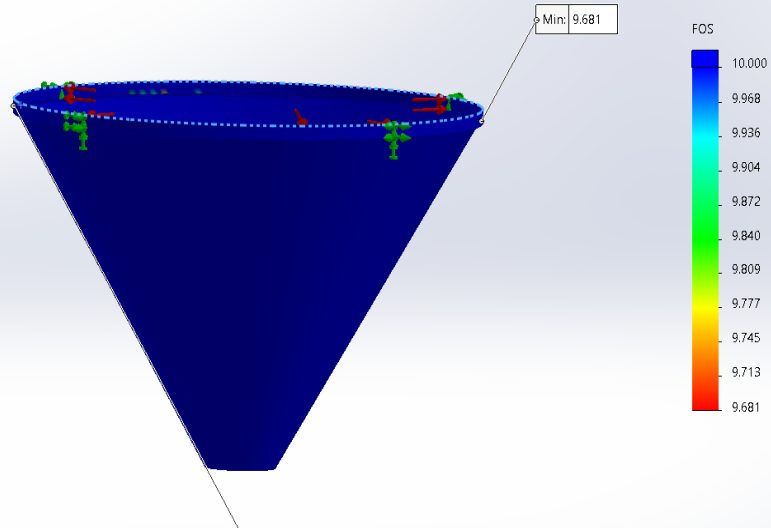
Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	2.548e-14 Element: 151331	5.027e-04 Element: 172683

Model name: auger and shaft
Study name: auger and shaft analysis(-Default-)
Plot type: Static strain Strain1
Deformation scale: 16,310.6



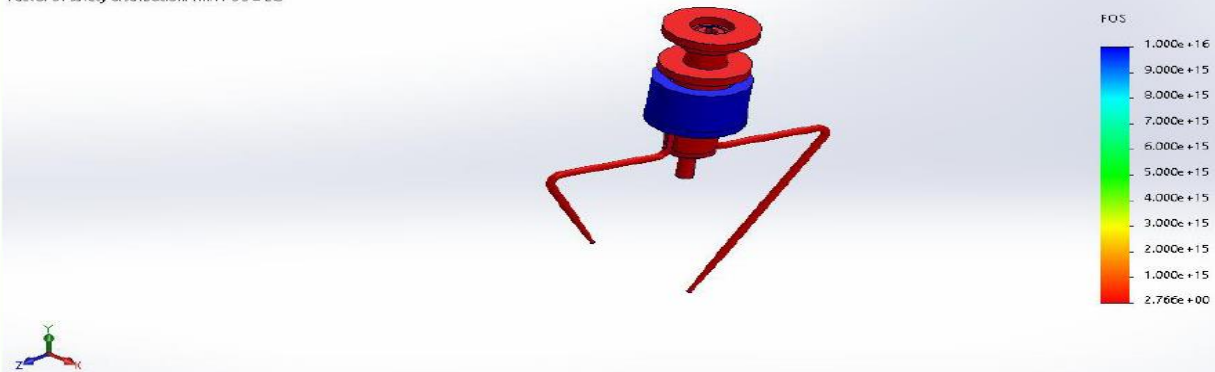


Model name: hopper2
Study name: Static 1(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion: Max von Mises Stress
Factor of safety distribution: Min FOS = 9.7



Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	2.766e+00 Node: 437692	1.000e+16 Node: 70847

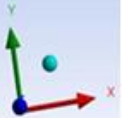
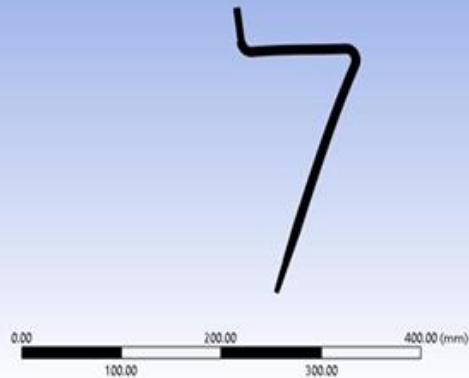
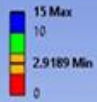
Model name: Assem4
Study name: assembly_1_static(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion: Max von Mises Stress
Factor of safety distribution: Min FOS = 2.8



Assem4-assembly_1_static-Factor of Safety-Factor of Safety1

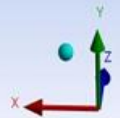
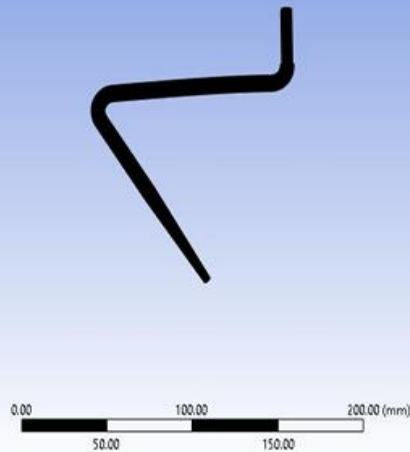
A: Static Structural
Safety Factor
Type: Safety Factor
Time: 1
2/24/2026 10:15 PM

Ansys
2023 R1



A: Static Structural
Safety Factor
Type: Safety Factor
2/26/2026 9:46 PM

Ansys
2023 R1



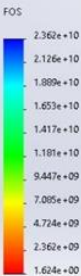
Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	2.281e+00 Node: 196616	6.471e+07 Node: 213371

Model name: auger and shaft and plate
Study name: auger and shaft and plate analysis(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Max von Mises Stress
Factor of safety distribution: Min FOS = 2.3



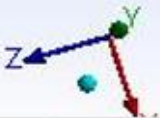
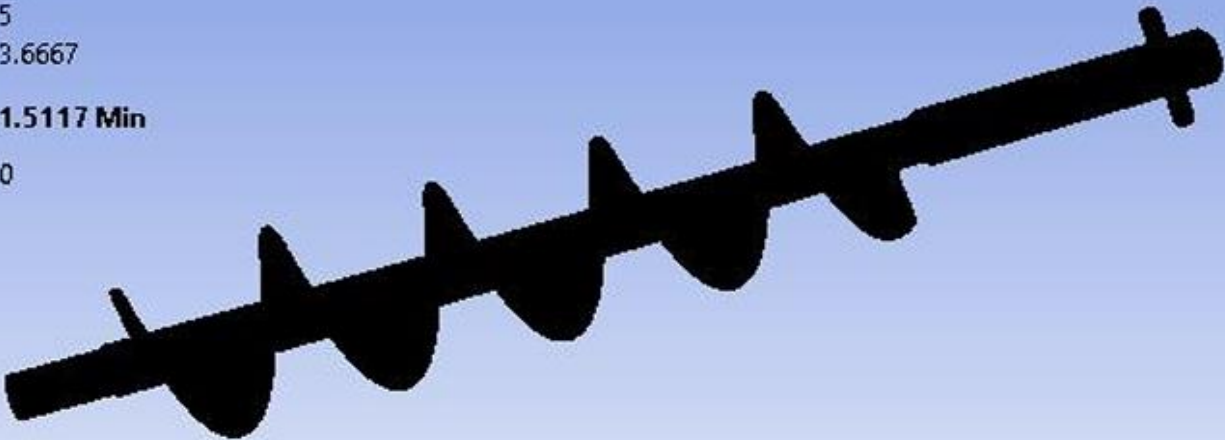
Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	1.624e+00 Node: 265549	2.362e+10 Node: 487491

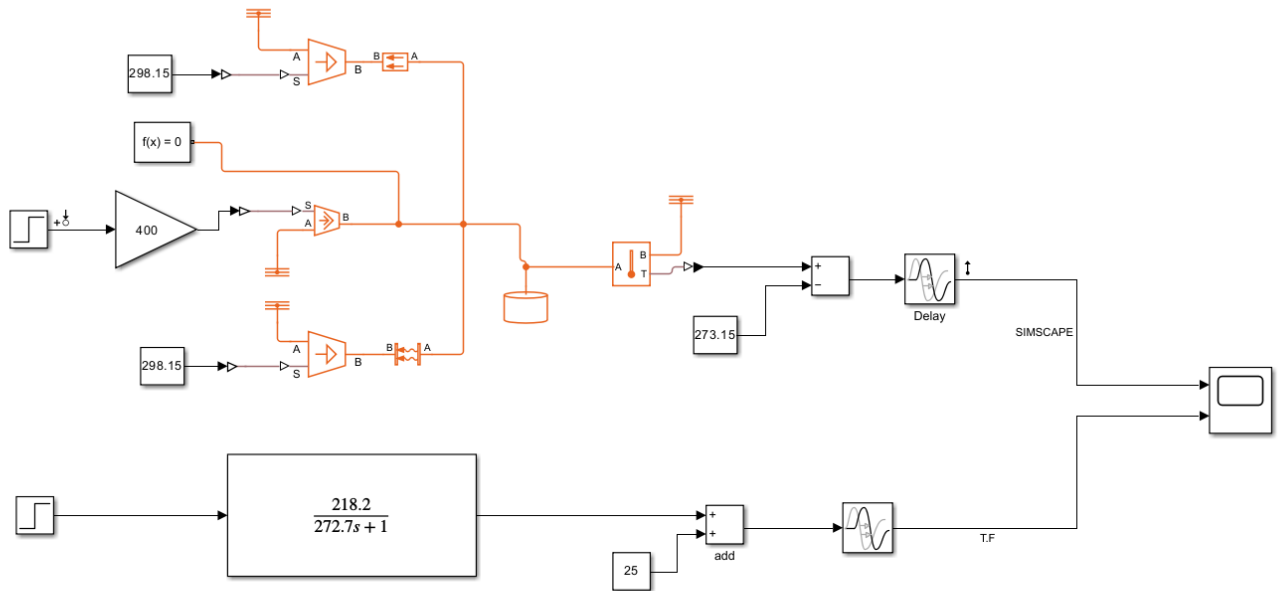
Model name: auger and shaft
Study name: auger and shaft analysis(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion: Max von Mises Stress
Factor of safety distribution: Min FOS = 1.6



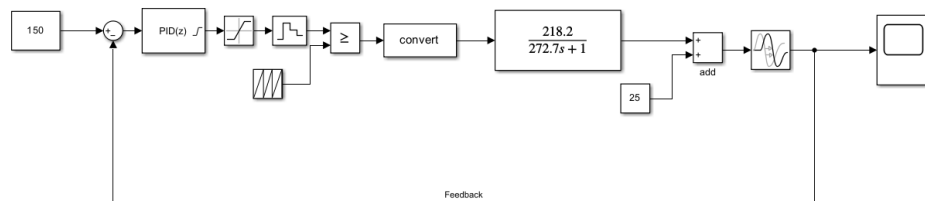
A: Static Structural
static factor
Type: Safety Factor
Time: 1
1/5/2026 2:56 AM

Ansys
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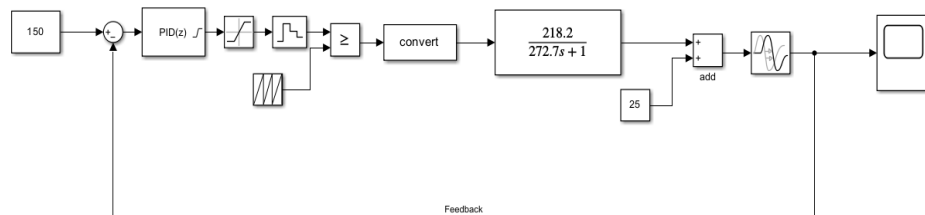




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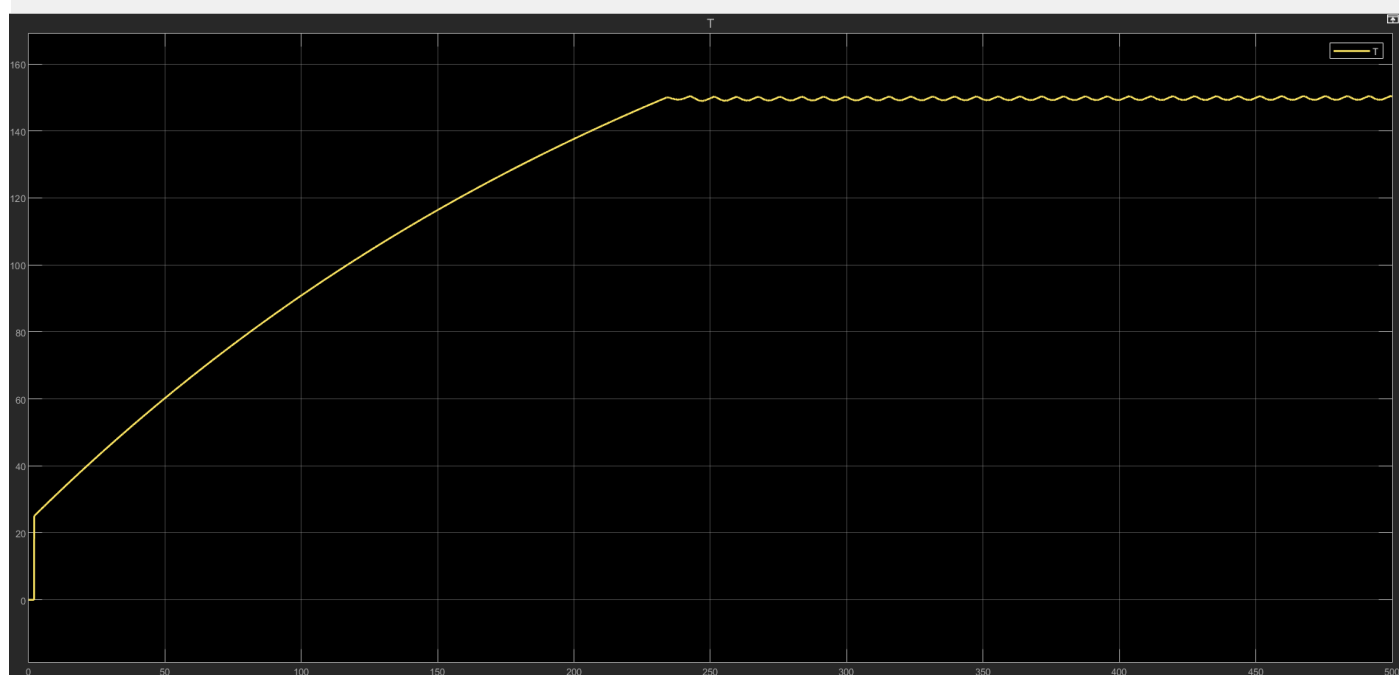
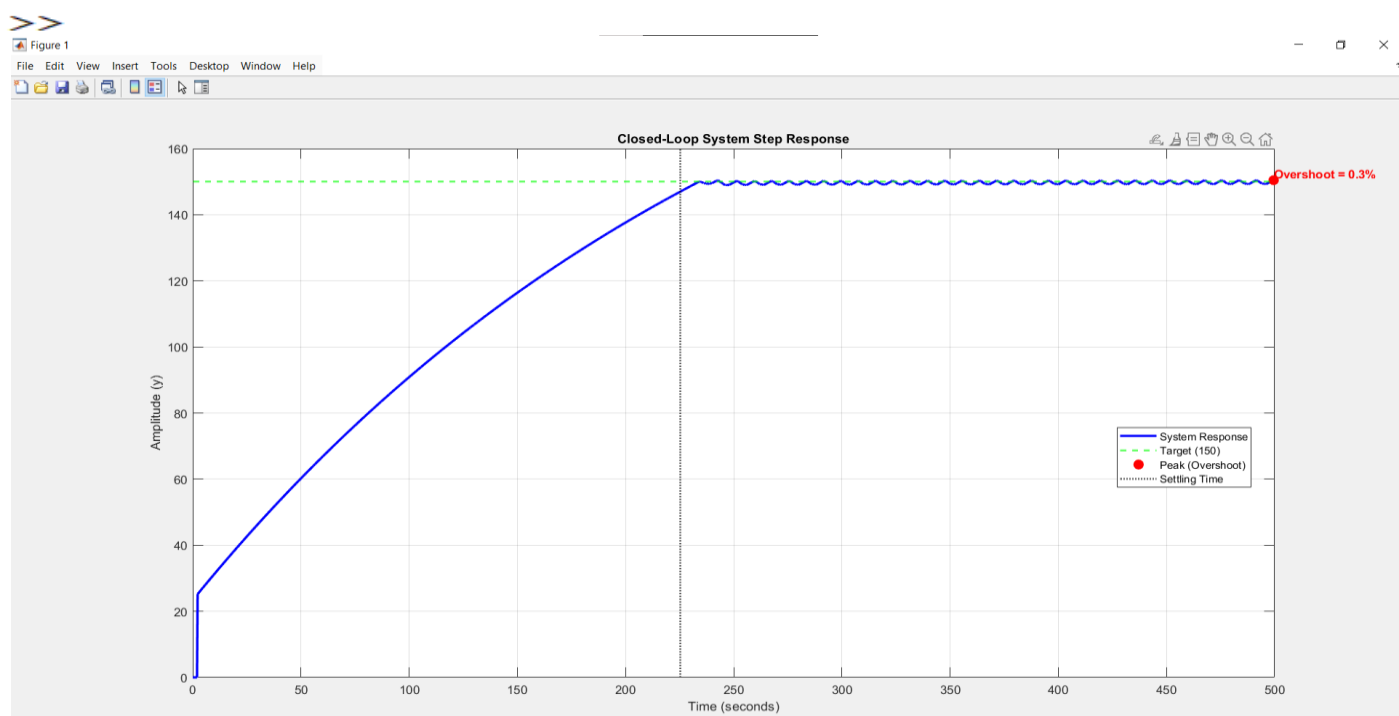


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>> t = out.t;  
>> info = stepinfo(y, t, 150);  
>> disp(info);
```

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TransientTime: 225.3366  
SettlingTime: 225.3366  
SettlingMin: 135.0039  
SettlingMax: 150.4561  
Overshoot: 0.3041  
Undershoot: 0  
    Peak: 150.4561  
PeakTime: 499.5200
```



Appendix D: Calculations and Tables

D.1. Motor Sizing Spreadsheet.

Optimal Motor Selection Table - VFFS Auger Filler

I. Basic Design Requirements

Item	Value	Unit	Notes
Required Design Torque (T_{design})	1.14	$N \cdot m$	With SF = 3.0
Design Torque with Reserve ($T_{design,total}$)	1.332	$N \cdot m$	$\times 1.2$ for unexpected conditions
Required Speed at Auger	150	RPM	For filling in 0.6 seconds
Maximum Angular Acceleration	300	rad/s^2	For rapid response

II. Comparative Motor Options

Specification	Non Acceptable Option	Acceptable Option	Current Option
Model	MSMF012L1U1	MSMF022L1U1	MSMF042L1U1
Manufacturer	Panasonic	Panasonic	Panasonic
Power	100 W	200 W	400 W
Rated Torque (T_{rated})	$0.32\ N \cdot m$	$0.64\ N \cdot m$	$1.27\ N \cdot m$
Peak Torque (T_{peak})	$0.95\ N \cdot m$	$1.91\ N \cdot m$	$3.82\ N \cdot m$

Specification	Non Acceptable Option	Acceptable Option	Current Option
Rated Speed	3000 RPM	3000 RPM	3000 RPM
Maximum Speed	6000 RPM	6000 RPM	6000 RPM
Rotor Inertia (J_{rotor})	$0.051 \times 10^{-4} \text{ kg} \cdot \text{m}^2$	$0.14 \times 10^{-4} \text{ kg} \cdot \text{m}^2$	$0.27 \times 10^{-4} \text{ kg} \cdot \text{m}^2$
Weight	0.47 kg	0.78 kg	1.2 kg
Flange Dimensions	40 mm	60 mm	60 mm
Protection Rating	IP67	IP67	IP67
Approximate Price	\$150	\$220	\$400

III. Gearbox Compatibility

Item	Non Acceptable Option	Acceptable Option	Current Option
Proposed Gear Ratio (i)	5:1	10:1	20:1 (Current)
Required Servo Torque	$1.48 \text{ N} \cdot \text{m}$	$0.74 \text{ N} \cdot \text{m}$	$0.37 \text{ N} \cdot \text{m}$
Available Servo Torque	$0.32 \text{ N} \cdot \text{m}$	$0.64 \text{ N} \cdot \text{m}$	$1.27 \text{ N} \cdot \text{m}$
Safety Margin (SF_{actual})	0.216	1	3.5

Item	Non Acceptable Option	Acceptable Option	Current Option
Evaluation	Not safe	Weak	Very safe
Speed at Auger	600 RPM	300 RPM	150 RPM
Required Modification	Electronic speed reduction	None	None

IV. Practical Application Compatibility

Criterion	Non Acceptable Option	Acceptable Option	Current Option (MSMF042L1U1)
Stopping Precision	High	High	Very High
Acceleration Response	Very Fast	Fast	Relatively Slow
Power Consumption	Low (30W)	Medium (60W)	High (120W)
Cooling Requirement	Natural	Natural	Additional Fan
Lifespan	Long (10+ years)	Very Long	Very Long
Maintenance Ease	Easy	Easy	Easy
Spare Parts Availability	Bad	Not good	Excellent

Comprehensive Torque Equations Table - VFFS Auger Filler

I. Fundamental Material Torques

No.	Torque Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
1	Material Weight Torque	$T_w = m \times g \times r_m$	m = Material mass (0.5 kg) g = Gravity acceleration (9.81 m/s²) $r_m = \text{Mean radius} = \frac{r_o + r_i}{2}$	0.093	$N \cdot m$	Basic torque required to lift the material.
2	Axial Friction Torque	$T_f = \mu \times m \times g \times r_m$	μ = Friction coefficient (0.3 for fine salt)	0.028	$N \cdot m$	Material resistance to movement.
3	Wall Friction Torque	$T_{wall} = \mu_w \times f_n \times r_o$	μ_w = Wall friction coefficient (0.2-0.4) $F_n = \text{Normal force} = m \times g \times \cos(\alpha)$ r_o = Outer radius (0.03 m)	0.015	$N \cdot m$	Friction of the auger flight against the tube wall.
4	Internal Shear Torque	$T_s = \tau \times A \times r_m$	τ = Material shear stress (100-500 Pa for salt) $A = \text{Shear area} = \pi \times D_s \times L_f$	0.05	$N \cdot m$	Material resistance to deformation.

II. Dynamic Torques

No.	Torque Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
5	Auger Inertia Torque	$T_{j, auger} = J_{auger} \times \alpha$	$J_{auger} = \frac{1}{2} \times m_{auger} \times (r_o^2 + r_i^2)$ $\alpha = \text{Angular acceleration} = \frac{\Delta \omega}{\Delta t}$	0.008	$N \cdot m$	Impactful during startup and stopping.
6	Material Inertia Torque	$T_{j, material} = J_{material} \times \alpha$	$J_{material} = \frac{1}{2} \times m \times (r_o^2 + r_i^2)$	0.004	$N \cdot m$	Accounts for the mass of the accelerating material.
7	Centrifugal Torque	$T_c = m \times w^2 \times r_m \times \mu \times r_o$	$\omega = \text{Angular velocity (rad/s)}$ $\omega = \frac{2\pi \times N}{60}$	0.012	$N \cdot m$	Increases significantly at high speeds.
8	Vibration Torque	$T_v = m_v \times A_v \times R_m$	$m_v = \text{Vibrator mass (0.1 kg)}$ $a_v = \text{Vibration acceleration (2-5 m/s}^2\text{)}$	0.006	$N \cdot m$	Applicable if a vibration motor is utilized.

III. Mechanical Torques

No.	Torque Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
9	Bearing Friction Torque	$T_{bearing} = \frac{1}{2} \times \mu_b \times F_r \times d_b$	μ_b = Bearing friction coefficient (0.001-0.005) F_r = Radial load (N) d_b = Bearing diameter (0.02 m)	0.0002	N·m	Very small but critical for precision.
10	Gear Loss Torque	$T_{gear,loss} = T_{in} \times (1 - \eta)$	T_{in} = Input torque η = Gear efficiency (0.94)	Variable	N·m	Represents roughly 6% of lost power.
11	Coupling Loss Torque	$T_{coupling} = \mu_c \times F_c \times r_c$	μ_c = Coupling friction coefficient (0.1) F_c = Coupling compression force r_c = Coupling radius	0.003	N·m	Minor losses occurring at the connections.
12	Air Resistance Torque	$T_{air} = \frac{1}{2} \times \rho_{air} \times c_d \times A \times v^2 \times r_o$	ρ_{air} = Air density (1.225 kg/m³) c_d = Drag coefficient (0.5-1.0) A = Exposed area v = Linear velocity	0.001	N · m	Extremely negligible, typically ignored.

IV. Final Design Torques

No.	Torque Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
13	Total Raw Torque	$T_{raw} = T_w + T_f + T_{wall} + T_s + T_{j,total} + T_C + T_{bearing} + T_{coupling}$	Sum of all preceding torques	0.37	$N \cdot m$	Value before applying any safety factors.
14	Required Design Torque	$T_{design} = T_{raw} \times SF$	SF = Safety Factor (3.0 as specified)	1.14	$N \cdot m$	The primary value used for design.
15	Design Torque with Reserve	$T_{design,total} = T_{design} \times 1.2$	1.2 = Reserve factor for unexpected operational conditions	1.37	$N \cdot m$	Ensures components are never overstressed.

V. Servo Conversion Equations

No.	Torque Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
16	Required Servo Torque	$T_{servo} = \frac{T_{design,total}}{i \times n}$	i = Gear ratio (20) n = Gear efficiency (0.94)	0.033	N·m	Torque required directly at the servo shaft.
17	Servo Peak Torque (Acceleration)	$T_{servo,peak} = T_{servo} + \frac{J_{total} \times \alpha_{max}}{i \times n}$	J_{total} = Total system inertia α_{max} = Maximum acceleration	0.045	$N \cdot m$	Torque spike during maximum acceleration.
18	Actual Safety Margin	$SF_{actual} = \frac{T_{servo,rated}}{T_{servo}}$	$T_{servo,rated}$ = Servo rated torque (1.27 N · m)	38.5	-	Excessively high (confirms the motor is oversized).
19	Gear Output Torque	$T_{output} = T_{servo,rate} \times i \times n$	-	23.9	$N \cdot m$	Total available torque at the auger shaft.

Comprehensive Volumetric Equations Table - Auger Filler VFFS

First: Basic Auger Geometry Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
1	Annular Cross-Sectional Area	$A = \frac{\pi}{4} \times (D^2 - D_s^2)$	D = Outer auger diameter (0.06 m) D _s = Central shaft diameter (0.016 m)	0.00267	m ²	Effective area for material transfer
2	Outer Radius	$r_o = \frac{D}{2}$	D = Outer diameter	0.03	m	For circular calculations
3	Inner Radius	$r_i = \frac{D_s}{2}$	D _s = Shaft diameter	0.008	m	For circular calculations
4	Mean Radius	$r_m = \frac{r_o + r_i}{2}$	-	0.019	m	For torque and force calculations
5	Radial Flight Height	$h_f = r_o - r_i = \frac{D - D_s}{2}$	-	0.022	m = 22 mm	Helical chamber depth
6	Outer Circumference	$C_o = \pi \times D$	-	0.1885	m	For linear speed calculations
7	Inner Circumference	$C_i = \pi \times D_s$	-	0.0503	m	For inner speed calculations

Second: Auger Volume Calculations Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
8	Theoretical Volume per Pitch	$V_{pitch,theory}$ $= A \times p$ $= \frac{\pi}{4} (D^2 - D_s^2) \times p$	$p =$ Pitch = Distance between flights (0.06 m)	0.000160	m ³ = 160 cm ³	Without considering voids
9	Effective Volume per Pitch	$V_{pitch,eff}$ $= \eta_v$ $\times V_{pitch,theory}$ $= \eta_v \times A \times p$	$\eta_v =$ Volumetric filling efficiency (0.6 for fine salt)	0.000096	m ³ = 96 cm ³	Considering voids between material particles
10	Total Auger Volume	$V_{auger,total}$ $= V_{pitch,eff}$ $\times n$ $= \eta_v \times A \times p$ $\times n$	n = Number of effective pitches (4.5)	0.000432	m ³ = 432 cm ³	Volume of material inside the auger
11	Volume per Complete Revolution	V_{rev} $= V_{pitch,eff}$ $= \eta_v \times A \times p$	-	0.000096	m ³	Same as Equation 9
12	Volume per Degree of Rotation	V_{deg} $= \frac{V_{pitch,eff}}{360^\circ}$ $= \frac{\eta_v \times A \times p}{360}$	-	2.67×10^{-7}	m ^{3/°}	For precise Servo control

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
13	Volume per Pulse	$V_{pulse} = \frac{V_{pitch,eff}}{PPR}$ $= \frac{\eta_v \times A \times p}{8,388,608}$	PPR = Encoder resolution (23-bit = 8,388,608)	1.14×10^{-11}	m ³ /pulse	Ultra-high precision control

Third: Mass & Flow Calculations Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
14	Mass per Pitch	$m_{pitch} = V_{pitch,eff} \times \rho_b$ $= \eta_v \times A \times p \times \rho_b$	ρ_b = Bulk density of material (1300 kg/m ³ for salt)	0.125	kg = 125 g	Theoretical mass per pitch
15	Mass per Revolution	$m_{rev} = m_{pitch}$ $= \eta_v \times A \times p \times \rho_b$	-	0.125	kg	Same as Equation 14
16	Total Auger Mass	$m_{auger,total} = V_{auger,total} \times \rho_b$ $= \eta_v \times A \times p \times n \times \rho_b$	-	0.562	kg = 562 g	Maximum mass inside the auger
17	Volumetric Flow Rate	$Q_v = V_{pitch,eff} \times N$ $= \eta_v \times A \times p \times N$	N = Angular speed (rev/sec)	Variable	m ³ /s	Material volume/time

No .	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
18	Mass Flow Rate	$m = \dot{Q}_v \times \rho_b$ $= \eta_v \times A \times p \times N \times \rho_b$	-	Variable	kg/s	Material mass/time
19	Target Dose Mass	$m_{dose,target} = 1.0$	Determined from customer requirements	1.0	kg = 1000 g	Target to be achieved
20	Required Revolutions per Dose	$N_{dose} = \frac{m_{dose,target}}{m_{rev}}$ $= \frac{m_{dose,target}}{\eta_v \times A \times p \times \rho_b}$	-	8.0	rev	To obtain 1000 g
21	Required Pitches per Dose	$n_{dose} = \frac{m_{dose,target}}{m_{pitch}}$ $= \frac{m_{dose,target}}{\eta_v \times A \times p \times \rho_b}$	-	8.0	pitch	Same as Equation 20
22	Required Rotation Angle	$\theta_{dose} = N_{dose} \times 360$ $= \frac{m_{dose,target} \times 360}{\eta_v \times A \times p \times \rho_b}$	-	2880	degree	For Servo Position Control
23	Required Filling Time	$\frac{N_{dose}}{N_{auger}}$ $= \frac{m_{dose,target} \times 360}{\eta_v \times A \times p \times \rho_b \times N_{auger}}$	N_{auger} = Auger speed (rev/sec)	0.6	s	For synchronization with bag cycle

Fourth: Spinner Plate Volume Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
24	Circular Plate Area	$A_{disc} = \frac{\pi}{4} \times D_{disc}^2$	D_{disc} = Plate diameter (0.10 m)	0.00785	m ²	Total area of the plate
25	Sloped Area	$A_{slop} = \pi \times (r_{disc}^2 - r_{flat}^2)$	r_{disc} = Plate radius (0.05 m) r_{flat} = Flat area radius (0.034775 m)	0.00415	m ²	Effective sliding area
26	Material Volume on Plate	$V_{disc} = A_{disc} \times h_{layer} = \frac{\pi}{4} \times D_{disc}^2 \times h_{layer}$	h_{layer} = Material layer thickness (0.01-0.02 m)	0.0000785	m ³ = 78.5 cm ³	For continuous feeding
27	Material Mass on Plate	$m_{disc} = V_{disc} \times \rho_b = A_{disc} \times h_{layer} \times \rho_b$	-	0.102	kg = 102 g	Reserve above the auger
28	Effective Slope Angle	$\alpha_{eff} = \arctan\left(\frac{h_{disc}}{r_{disc} - r_{flat}}\right)$	h_{disc} = Plate edge height (0.01 m)	55	degree	Affects flow speed

Fifth: Hopper Volume Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
29	Cylindrical Hopper Volume	$V_{hopper,cyl} = \frac{\pi}{4} \times D_h^2 \times H_h$	D_h = Hopper diameter (0.3 m) H_h = Hopper height (0.4 m)	0.0283	m ³ = 28.3 L	For temporary storage
30	Conical Hopper Volume	$V_{hopper,cyl} = \frac{\pi}{12} \times H_c \times (D_t^2 + D_t \times D_b + D_b^2)$	H_c = Cone height D_t = Top diameter D_b = Bottom diameter	Variable	m ³	For smooth feeding
31	Material Volume in Hopper	$V_{material,hopper} = \eta_{fill} \times V_{hopper}$	η_{fill} = Hopper fill factor (0.8-0.9)	Variable	m ³	Less than geometric volume
32	Material Mass in Hopper	$m_{hopper} = V_{material,hopper} \times \rho_b$	-	Variable	kg	For production planning
33	Complete Emptying Time	$t_{empty} = \frac{m_{hopper}}{\dot{m}} = \frac{V_{material,hopper}}{Q_v}$	-	Variable	s	Between two hopper refills

Sixth: Bag Volume Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
34	Rectangular Bag Volume	$V_{bag} = \omega \times h \times t$	ω = Width (0.15 m) h = Height (0.20 m) t = Thickness/Depth (0.08 m)	0.0024	m ³ = 2.4 L	Internal bag volume
35	Available Product Volume	$V_{product} = V_{bag} \times \eta_{fill,bag} = \omega \times h \times t \times \eta_{fill,bag}$	$\eta_{fill,bag}$ = Bag fill factor (0.7-0.8)	0.0018	m ³ = 1.8 L	Considering air
36	Maximum Bag Mass	$m_{bag,max} = \rho_b \times V_{product}$	-	2.34	kg	For safety verification
37	Required Film Length	$L_{film} = 2 \times (\omega + t) + L_{seal}$	L_{seal} = Seal margin (0.02 m)	0.48	m = 480 mm	Per bag
38	Required Film Width	$w_{film} = h + L_{seal,top} + L_{seal,bottom}$	$L_{seal,bottom}$ = Top/bottom seal margin	0.24	m = 240 mm	Roll width

Seventh: Pressure & Stress Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
39	Hydrostatic Column Pressure	$P_h = \rho_b \times g \times h_{\text{column}}$	h_{column} = Material column height above auger (0.2 m)	2550	Pa = 2.55 kPa	Pressure on the auger
40	Vertical Force from Pressure	$F_h = P_h \times A = \rho_b \times g \times h_{\text{column}} \times \frac{\pi}{4} \times D_{\text{disc}}^2$	-	20.0	N	Needed for structural design
41	Shear Stress in Material	$\tau = \frac{F_{\text{shear}}}{A_{\text{shear}}} = \frac{T}{2\pi \times r_m^2 \times h_f}$	T = Applied torque	Variable	Pa	For viscous materials
42	Compressibility Factor	$K = \frac{\Delta P}{\frac{\Delta V}{V_o}}$	ΔP = Pressure change ΔV = Volume change	Variable	Pa	For compressible materials

Eighth: Speed & Time Equations

No.	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
43	Angular Velocity	$\omega = 2\pi \times N$ $= \frac{2\pi \times n_{RPM}}{60}$	N = Speed (rev/sec) n_{RPM} = Speed (rev/min)	15.7	rad/s	For unit conversion
44	Linear Material Velocity	$v_{material} = \omega \times r_m \times \tan(\phi)$	ϕ = Internal friction angle (25-35°)	Variable	m/s	Actual transfer speed
45	Complete Filling Cycle Time	$T_{cycle} = t_{fill} + t_{settle} + t_{seal} + t_{cut} + t_{frad}$	t_{settle} = Settling time t_{seal} = Sealing time t_{cut} = Cutting time t_{frad} = Feeding time	1.71	s	For total productivity
46	Productivity	$\dot{n} = \frac{60}{T_{cycle}}$	-	35	bag/min	Production target

Ninth: Correction & Calibration Equations

No .	Equation Name	Equation	Symbols & Meanings	Calculated Value	Unit	Engineering Notes
47	Volumetric Correction Factor	$C_v = \frac{m_{actual}}{m_{theory}}$ $= \frac{m_{actual}}{\eta_v \times A \times p \times \rho_b \times n}$	m_{actual} = Measured actual mass	Variable	-	Determined experimentally
48	Actual Bulk Density	$\rho_{actual} = \frac{m_{actual}}{V_{auger, total}}$ $= \frac{m_{actual}}{\eta_v \times A \times p \times n}$	-	Variable	kg/m ³	May differ from 1300
49	Actual Filling Efficiency	$\eta_{actual} = \frac{m_{actual}}{\rho_b \times A \times p \times n}$	-	Variable	-	For precise calibration
50	Corrected Number of Revolutions	$N_{corrected} = N_{theory} \times \frac{1}{C_v}$ $= \frac{m_{target}}{m_{actual}} \times N_{theory}$	-	Variable	rev	For precise control

Summary of Key Values

Item	Value	Unit	Main Equation
Annular Area (A)	0.00267	m^2	$A = \frac{\pi}{4} \cdot (D^2 - D_s^2)$
Effective Volume per Pitch	96	cm^3	$\eta_v \times A \times p$
Total Auger Volume	432	cm^3	$\eta_v \times A \times p \times n$
Mass per Pitch	125	g	$\eta_v \times A \times p \times \rho_b$
Total Auger Mass	562	g	$\eta_v \times A \times p \times \rho_b \times n$
Required Revolutions for 1000g	8.0	rev	$\frac{m_{target}}{\eta_v \times A \times p \times \rho_b}$
Rotation Angle for 1000g	2880	degree	$N_{dose} \times 360$
Filling Time	0.6	s	$\frac{N_{dose}}{N_{auger}}$

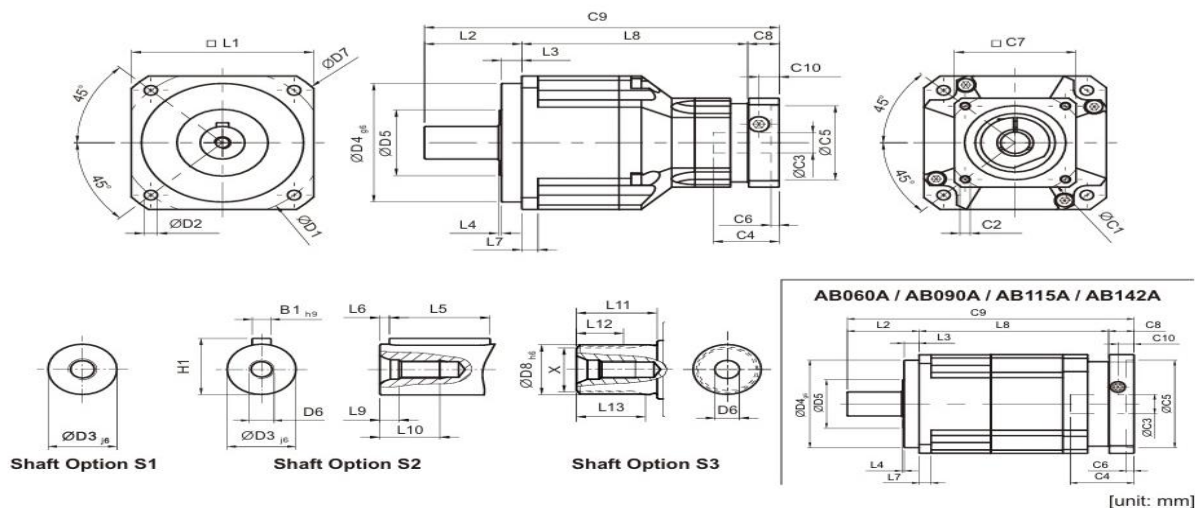
Required Design Torque (T_{design})	1.14	$N \cdot m$	With SF = 3.0
Design Torque with Reserve ($T_{design,total}$)	1.332	$N \cdot m$	$\times 1.2$ for unexpected conditions
Required Speed at Auger	150	RPM	For filling in 0.6 seconds
Maximum Angular Acceleration	300	rad/s^2	For rapid response

Spec Detail

Item	Specifications
Part Number	MSMF042L1U1
Details	Low inertia, Connector type
Family Name	MINAS A6
Series	MSMF Series
Type	Low inertia
Protection class	IP67
About Enclosure	Except rotating portion of output shaft and connecting pin part of the motor connector and the encoder connector.
Environmental Conditions	For more details, please refer to the instruction manual.
Flange sq. dimension	60 mm sq.
Flange sq. dimension (Unit:mm)	60
Motor lead-out configuration	Connector
Motor encoder connector	Connector
Power supply capacity (kVA)	0.9
Voltage specifications	200 V
Rated output	400 W
Rated current (A (rms))	2.4
Holding brake	without
Mass (kg)	1.2
Oil seal	with
Shaft	Key-way, center tap
Rated torque (N · m)	1.27
Continuous stall torque (N · m)	1.27

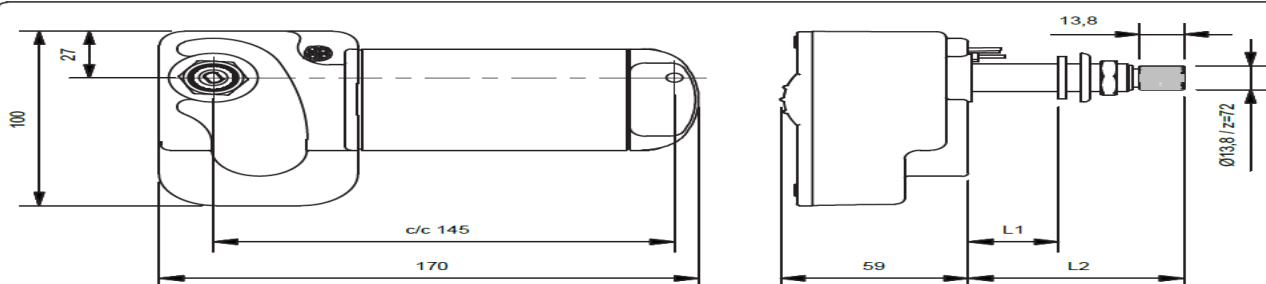
Part Number	MBDLN25SE
Details	A6 SE series Position control type (Incremental system only, Pulse train only) without the safety function
Family Name	MINAS A6
Series	A6 SE series
Type	Position control type
Frame	B-Frame
Frequency response	3.2 kHz
Control method	Position control
About control method	A6 SE series driver (Position control only) does not correspond to the absolute system of using the serial communication with the host device. It supports incremental system only.
Safety Function	without
Supply voltage	Single/3-phase 200 V
I/F Classification of type	Pulse train only
Dimensions (W) (Unit: mm)	55
Dimensions (H) (Unit: mm)	150
Dimensions (D) (Unit: mm)	130
Mass (kg)	1.0
Environment	For more details, please refer to the instruction manual.

Dimensions (2-stage, Ratio i=12~100) / AB Series



Dimension	AB042	AB060	AB060A	AB090	AB090A	AB115	AB115A	AB142	AB142A	AB180	AB220
D1	50	70		100		130		165		215	250
D2	3.4	5.5		6.6		9		11		13	17
D3 _{j6}	13	16		22		32		40		55	75
D4 _{g6}	35	50		80		110		130		160	180
D5	22	45		65		95		75		95	115
D6	M4x0.7P	M5x0.8P		M8x1.25P		M12x1.75P		M16x2P		M20x2.5P	M20x2.5P
D7	56	80		116		152		185		240	292
D8 _{h6}	-	16		22		32		40		55	75
L1	42	60		90		115		142		180	220
L2	26	37		48		65		97		105	138
L3	5.5	7		10		12		15		20	30
L4	1	1.5		1.5		2		3		3	3
L5	16	25		32		40		63		70	90
L6	2	2		3		5		5		6	7
L7	4	6		8		10		12		15	20
L8	58.5	72	98	111.5	126.5	143.5	163	176	191	209.5	248
L9	4.5	4.8		7.2		10		12		15	15
L10	10	12.5		19		28		36		42	42
L11	-	26		26		26		40		41.5	52
L12	-	15		15		15		20		21.5	28
L13	-	21		22.5		23		33.5		33.5	45
C1 ³	46	46	70	70	100	100	130	130	165	165	215
C2 ³	M4x0.7P	M4x0.7P	M5x0.8P	M5x0.8P	M6x1P	M6x1P	M8x1.25P	M8x1.25P	M10x1.5P	M10x1.5P	M12x1.75P
C3 ³	≤11 / ≤12	≤11 / ≤12	≤14 / ≤16	≤14 / ≤15.875 / ≤16	≤19 / ≤24	≤19 / ≤24	≤32	≤32	≤38	≤38	≤48
C4 ³	25	25	34	34	40	40	50	50	60	60	85
C5 ³	30	30	50	50	80	80	110	110	130	130	180
C6 ³	3.5	3.5	8	8	4	4	5	5	6	6	6
C7 ³	42	42	60	60	90	90	115	115	142	142	190
C8 ³	29.5	29.5	19	19	17	17	19.5	19.5	22.5	22.5	29
C9 ³	114	138.5	154	178.5	191.5	225.5	247.5	292.5	310.5	337	415
C10 ³	8.75	8.75	13.5	13.5	10.75	10.75	13	13	15	15	20.75
B1 _{h9}	5	5		6		10		12		16	20
H1	15	18		24.5		35		43		59	79.5
X	-	W16x0.8x30x18x6m		W22x1.25x30x16x6m		W32x1.25x30x24x6m		W40x2x30x18x6m		W55x2x30x26x6m	W70x2x30x34x6m
DIN5480	-										

3. C1-C10 are motor specific dimensions (metric std shown). Refer to www.apexdyna.com and Design Tool to view your specific motor mounting system.



Variations:

Max length:

	L2	12 V	24 V
16	43	533011	533012
42	68	533021	533022
68	94	533031	533032

Current diagram:

	12 V	24 V
No load	< 2 A	< 1 A
Rated	< 4 A	< 2 A
Stall	< 9 A	< 4.5 A
Fuse (Fast blow)	5 A	3 A
Speed no load	(35 / 55 RPM)	

Limited warranty

- Roca warrants the wiper motors to be free from defective material and workmanship for a period of 3 years from the date of purchase.
- This warranty does not apply to failure or damage caused by improper installation, misuse, lightning strike or damage from a power surge.
- If failure occurs during this period, return it to your dealer for repair or replacement.

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Go to Se

D.2. Bearing and Fastener Selection Tables.

Configured Specifications of 6809ZZ

Inner Dia. d	45	Outer Dia. D	58
Width B	7	Raceway Ring Shape	Ball
Bearing Type	Double Shielded	Outer Ring Type	Flat
Inner/Outer Ring Material	Steel	Basic Dynamic Load Rating	5350
Specifications/Environment	Standard	Precision (JIS)	Grade 0
Load Direction	Radial / Thrust	Rolling Element Material	Steel
Size Standards	Metric System	Number of Raceway Ring Rows	Single Row
Max. Allowable Rotational Speed	11000	Retainer Type	Punching

Configured Specifications of 6804ZZ

Inner Dia. d	20	Outer Dia. D	32
Width B	7	Raceway Ring Shape	Ball
Bearing Type	Double Shielded	Outer Ring Type	Flat
Inner/Outer Ring Material	Steel	Basic Dynamic Load Rating	4000
Specifications/Environment	Standard	Precision (JIS)	Grade 0
Load Direction	Radial / Thrust	Rolling Element Material	Steel
Size Standards	Metric System	Number of Raceway Ring Rows	Single Row
Max. Allowable Rotational Speed	21000	Retainer Type	Punching

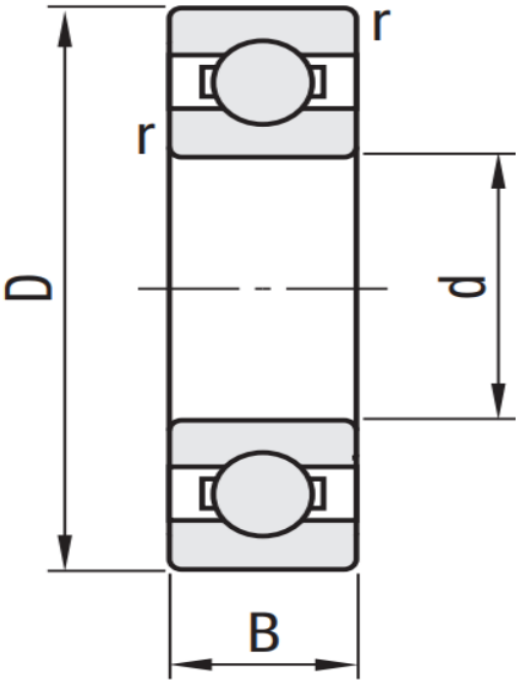
Configured Specifications of MUEN15

Sliding Surface	Shaft	Sealed Substance	Oil / Grease
Shaft Dia. d (s.d)	15	Oil Seal Material	Nitrile Rubber (NBR Nitrile Rubber)
Seal Rim Area Material	Outer Rubber Type	Environment, Applications	Standard
Dust Lip	Yes	Attaching Method	Only from shaft end
Housing Groove	Not Provided	Two Liquids Separation	No
Operating Temperature Range	-25 ~ 120	Operating Pressure	~ 0.03
RoHS	10	-	-

Part Number	d	D	B	r (min)	Basic Load Rating		Allowable Rotational Speed rpm (Reference)	Relative Dimensions				Mass (g) (Reference)
					Cr(Dynamic) kN	Cor(Static) kN		Ds		dh (max)	R (max)	
								(min)	(max)			
B6700ZZ		15	4	0.2	0.85	0.43	15000	10.4	11.21	14.2	0.2	2

6902 Ball Bearings 15X28X7mm Specifications & Dimensions

Universal	6902
Type	Ball
Inner/Outer Ring Material	Steel
Precision	Grade 0
Inner Dimension d(Ø)	15
Outer Dimension D(Ø)	28
Width B (or T)(mm)	7
Retainer Type	Punching
Load Direction	Radial
Specifications/Environment	Standard
Number of Raceway Ring Rows	Single Track
Size Standards	Metric Series



Configured Specifications of C-HTPA20S5M150-A-P15

Type	Timing Pulleys	Belt Type	S5M
Belt Width Used	15	No. of Teeth	20
Pulley Shape	Shape A	Material	[Aluminum] Aluminum Alloy
Surface Treatment	Clear Anodize	Timing Pulley / Idler	Pulleys
Shaft Bore Dia. d	15	Shaft Bore Dia. D	-
Shaft Hole Shape	P	RoHS	10



Mouse over to zoom



Oil Free Seals for Rotary Motion, Double Lip

Part Number: **MUEN45** [Clear](#)

- Shaft Operation: Rotation
- Adhesive Material: Nitrile Rubber...

Configured Specifications [↓](#)

Sliding Surface	Axis
Sealed Substance	Oil / Grease
Oil Seals material	Nitrile Rubber (NBR)
Shaft Diameter d (s.d.)(Ø)	45
Two Liquids Separation	No
Dust Lip	Presence
RoHS	10

Housing Groove	No
Operating Temperature Range(°C)	-25 ~ 120
Seal Rim Area Material	Outer Rubber Type
Attaching Method	From only the shaft end
Environment, Applications	Standard
Operating Pressure(MPa)	~ 0.03
-	-

II. Controller Selection (ESP32 vs. PLC)

Feature	Custom ESP32 Architecture	Traditional Industrial PLC
Processing Power	Dual-Core 240 MHz (Parallel)	Single-Core 50-100 MHz (Cyclic)
Motion Control	Native RMT (Hardware-level)	Needs High-Speed IO Modules
Cost Efficiency	Highly Cost-Effective	High License & Hardware Costs
Connectivity	Native WiFi/UART (Industry 4.0)	Requires Communication Gateways

Industrial Hardware Specifications

Component	Part Number/Type	Technical Justification
Thermal Sensor	K-Type + MAX6675	High range (up to 1024 ^{°C}) & SPI Digital interface.
Eye-mark Sensor	KS-C2 (NPN)	High-speed contrast detection for registration.
Heater Relay	Zero-Crossing SSR 40A	Eliminates EMI noise during PID switching.
Logic Isolation	Optocoupler Relay Module	Protects ESP32 (3.3V) from 24V industrial spikes.
Power Supply	MeanWell 24V, 14.6A	Provides stable power for motors and actuators.

D.3. Error Analysis Calculations.

II. Simulation vs. Requirement Analysis

Metric	Design Requirement	Simscape Result	Percentage Error / Status
Setpoint	150	150	0% (Exact match)
Settling Time	300	225.3 s	Within Bounds (PASSED)
Max Overshoot	< 1\%	0.304\%	Highly Stable (PASSED)
Steady-State Error	1	0.45	Improved precision
Derivative Filter (N)	Stable ()	1-NT_s	<1\$)